

Trading suspensions and food price inflation

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Abstract

Since the global food price inflation of 2008-09, concerns regarding futures market speculation and its impact on food prices have grown. Conflicting results from existing studies highlight the limitation of the research design. We apply the synthetic control methodology to examine if futures trading suspensions bring down food prices. Analysing three recent agri-futures trading suspensions events from India, we find no empirical evidence that such suspensions impact food prices. The counterfactual for each episode indicates that even in the absence of such suspensions, food prices would have followed a similar trend. Our findings inform the policy debate on agricultural market interventions related to derivatives trading.

Keywords: food price inflation, agricultural derivatives, trading suspensions, synthetic control, India

JEL Classification Codes: Q13, Q18, G1, G18

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1 Introduction

Food price inflation is a global concern. High prices have implications on food security, overall inflation, and can cause social unrest (Bellemare, 2015). Thus, countries across the world intervene in their food markets to ensure price stability. These interventions typically take the form of import relaxations, export curbs, and release of reserve stocks (Stucchi *et al.*, 2018; Abbott, 2012; Demeke *et al.*, 2012; Jones and Kwieciński, 2010) to ensure adequate food supply in domestic markets.

Since the global food price inflation of 2007-08, an added concern has been the role of excessive speculation in derivatives market and its impact on food prices. Between 2003 and 2008, institutional investments in commodity futures markets were estimated to exceed USD 200 billion (CFTC, September 2008). The increasing financialization of commodity markets, coinciding with rising global food prices, raised concerns that futures market speculation was distorting spot food prices (Cheng and Xiong, 2014; Masters, M., 2008).

Similar concerns were raised in India where national commodity derivatives exchanges were set up as recently as 2003 to aid efficient price discovery and effective risk management. However, with the concern that excessive speculation on commodity futures market exacerbated food price inflation and volatility, in 2007-08, India suspended derivatives trading in eight agricultural commodities traded on its national commodity futures exchanges.¹ As prices cooled down, the suspension on some of these commodities was lifted but continued on other ‘sensitive’² commodities. Since then, lack of empirical evidence on the impact of agricultural derivatives on food inflation has prompted the regulators to suspend agricultural derivatives trading as and when there is a large surge in food prices.³

While several studies in the global context have examined the impact of financialization of commodity markets on food prices, the evidence is mixed and conflicting (Basak and Pavlova, 2016). Studies such as Gilbert (2010) and Singleton (2014) argue that non-information based institutional investor flows impacted commodity prices during the 2008-09 boom. Others such as Brunetti and Buyuksahin (2009), Stoll and Whaley (2010), Hamilton and Wu (2015) and Chari and Christiano (2017) find little evidence to support the claim. Cheng and Xiong (2014) attribute these conflicting findings to methodological limitations. In particular, they note that approaches based on contemporaneous correlations, cross-sectional comparisons, or Granger causality often lack the power to identify causal impacts. The lack of a clear identification strategy can lead to erroneous inferences, an issue that the modern causal inference methods in the spirit of Angrist and Pischke (2010) can help address. It is in this context that we revisit the question of whether derivatives trading exacerbates food price inflation.

¹In the United States, the Onion Futures Act of 1958 prohibits trading in onion futures contracts. The Act remains in effect to this day.

²These included wheat, rice, sugar and some pulses including *tur* and *urad*.

³The latest episode was in December 2021 when the regulator suspended seven commodities from derivatives trading amidst high food prices. The suspension remains in place as of August 2025. See Appendix Table A.1 for a list of agricultural derivatives suspensions announced since the launch of national commodity derivatives exchanges in India.

Specifically, we ask: does derivatives trading suspension rein in food price inflation? What would have been the price trajectory in the absence of such suspensions? To answer these questions, we draw on the recent advances in the policy evaluation literature. In particular, we use the synthetic control methodology developed by Abadie *et al.* (2015), Abadie *et al.* (2010), and Abadie and Gardeazabal (2003) to identify the causal impact of agricultural derivatives trading suspensions on food prices. The methodology enables us to construct the counterfactual, referred to as the synthetic control, using a weighted combination of unaffected comparison units. By comparing the actual price trajectory for the treated unit with that of the synthetic control, we assess whether the observed price path would have been different in the absence of derivatives trading suspension.

The synthetic control methodology offers several advantages over the traditional regression approach (Abadie, 2021), which is not well suited for case studies when an intervention is implemented at an aggregate level, affecting a small number of large units (e.g. regions, countries), or a single unit, as in the present study. Regression analysis typically requires large samples and multiple instances of the event, making it ill-suited for estimating the effects of infrequent events. Moreover, unlike a regression, the synthetic control methodology explicitly assigns weights to each comparison unit and avoids the risks of extrapolation. The methodology has been used in several contexts, including studies on the effects of fintech on poverty alleviation (Adu and Hartarska, 2025), law and crime (Donohue *et al.*, 2019), fiscal rules on budget outcomes (Eliason and Lutz, 2018), and export restrictions on domestic production (Stucchi *et al.*, 2018), among others.

We examine the most recent episodes of agricultural derivatives trading suspension in India, implemented in response to rising food prices. These are: the August 2021 suspension of Gram / Bengal Gram (*chana*) derivatives in the pulses category and the October 2021 suspension of mustard seed derivatives in the oilseeds category. In addition, we also analyse an earlier suspension episode of *chana* derivatives in June 2016. Both *chana* and mustard seed contracts are among the most actively traded agricultural derivatives in India and therefore serve as a useful case study to analyse if derivatives suspension rein in food price inflation. Further, unlike other episodes of derivatives suspension where multiple commodities were suspended simultaneously, leaving no control units to construct the synthetic control, the events we analyse are single commodity suspension episodes. This setting enables us to use the other unaffected commodities as controls to identify the impacts, if any.

Using data on prices as well as predictor variables including international prices, global production shocks, domestic production, net imports and arrivals,⁴ we construct a synthetic control for each suspension episode to examine if derivatives suspension had any effect on food price inflation. Our results provide no empirical evidence that the suspensions had any impact on food prices. The analysis shows that prices of both commodities, *chana* and mustard oil, would have followed similar trajectories even without the suspension. The presence of the same trend between the treated series and their synthetic counterparts suggests that even in the absence of derivatives suspension, prices

⁴Arrivals refers to the quantity of produce brought to a regulated wholesale market (*mandi*) for sale on a given day.

would have evolved similarly. ‘In-space’ and ‘in-time’ placebos further confirm our findings. Our results remain robust to sensitivity tests based on alternative choices of predictor variables as well as ‘leave one out’ distribution of the synthetic control.

To explain our findings, we analyse derivative trading volumes for both commodities prior to the suspension and examine their association with spot market food prices using Granger causality tests. We find no evidence that changes in derivatives trading volumes led spot price changes during the pre-suspension period. We also observe that price trends in both *chana* and mustard, before and after the suspension, closely tracked those of other commodities in the same commodity group (oilseeds and pulses), irrespective of their trading status in derivatives markets. Structural break tests on domestic mustard oil prices and its international benchmark, canola futures, further show that domestic prices followed global trends. Taken together, these findings suggest that other factors, rather than derivatives trading activity, drove the observed price increases.

Our findings have important policy implications. A well-functioning agricultural derivatives market enables efficient price discovery, facilitates risk management, and supports farmers and processors in making production, storage and marketing decisions. Its role becomes particularly critical during periods of heightened price volatility. In the absence of empirical evidence on the effects of agricultural derivatives suspensions, such interventions have often been employed as precautionary measures. Our study addresses this gap by providing causal evidence and informs the policy debate on interventions related to agricultural derivatives trading.

The paper contributes to the literature on the financialization of commodity markets by addressing the identification challenges that have limited prior empirical work on the price effects of derivatives trading. To the best of our knowledge, this is the first study to apply the synthetic control methodology to assess the causal impact of derivatives trading suspensions on food price behaviour.

More broadly, the study contributes to the growing literature on the impact of government interventions on food prices. Using the synthetic control approach, Stucchi *et al.* (2018) examine the effects of export restrictions on domestic production in Bolivia, and find that such controls shift supply and negatively impact domestic production. Studies such as Baylis *et al.* (2014), Saini and Gulati (2017), Narayanan and Tomar (2023) and Saroj *et al.* (2021) explore trade and marketing restrictions in India and find that such interventions lead to inefficiencies in the functioning of markets. Saini and Gulati (2017) document evidence of price distortion in Indian agriculture due to trade policies and argue that such policies have been subservient to the overall goal of ensuring food security to poor consumers. Baylis *et al.* (2014) find that India’s export ban of wheat and rice in 2007 resulted in domestic markets becoming disconnected from the world market and reduced market integration between producing and consuming regions within the country. Narayanan and Tomar (2023) show that the price deficiency payments implemented in one of the major states in India, tended to depress agricultural prices, potentially leading to a temporary supply glut. Our study complements this literature by evaluating an underexplored yet frequently used policy tool: the suspension of agricultural derivatives trading.

The rest of the paper is organised as follows: Section 2 provides a brief background on agricultural commodity derivatives in India along with detailed description of the two commodities analysed in this study. Section 3 discusses the data and the empirical strategy. Section 4 presents the results followed by a discussion in Section 5. Section 6 tests the sensitivity of our main findings using various robustness tests. Section 7 concludes the paper.

2 Background

2.1 Agricultural commodity derivatives market in India

The first futures trade in India was recorded in 1875 in cotton with the setting up of the Bombay Cotton Trade Association Ltd. (Bhattacharya, 2007). Subsequently, several exchanges came up in different parts of the country for futures trade in various commodities. However, frequent restrictions and suspensions by authorities resulted in low depth and poor liquidity in these markets (Sahadevan, 2002). The first major boost to the development of commodity derivatives market came in 2000 when the government announced liberalisation of these markets under the ‘National Agricultural Policy’. The new policy led to the setting up of a regulated organised national commodities derivatives exchanges that allowed participants to manage price risk and facilitate price discovery using futures contracts.

At present, there are five national exchanges in India that trade agriculture commodity derivative contracts. These are: Multi Commodity Exchange Limited (MCX), National Commodity & Derivative Exchange Limited (NCDEX), Bombay Stock Exchange Limited (BSE), National Stock Exchange Limited (NSE) and Indian Commodity Exchange Limited (ICEX).⁵ Of these, the MCX has a dominant share in trading of non-agricultural commodities such as crude oil, gold and metals, while the NCDEX has a dominant share in trading of agricultural commodities.

However, despite the National Agricultural Policy that recommended development of active commodities futures trading market, these markets continue to be subjected to frequent restrictions and commodity suspensions (Gulati *et al*, 2017). Table A.1 reports derivatives suspensions announced since the 2000s. As can be seen from the table, some of the commodities have been suspended more than once in the last two decades. Others like *tur* and *urad* have been suspended for more than a decade now.

In this paper we analyse the derivatives suspension of two major commodities: *chana* (gram / Bengal gram) in the pulses category, and mustard seed in the oilseeds category. We select these commodities for two main reasons. First, both had a significant share⁶ in agricultural commodity derivatives trading in India. Second, the suspensions in these cases were implemented as single-commodity interventions, enabling a clearer identification of causal effects.⁷ Below, we provide a

⁵Both NSE and BSE introduced commodity derivatives trading in 2018.

⁶Shares in 2020-21 for *chana* and mustard seed were 9.4 percent and 9.6 percent, respectively of the overall agricultural derivative trade in India.

⁷This contrasts with the suspensions implemented in early 2007, 2008, and December 2021, when four to six commodities were suspended simultaneously (see Table A.1). Such episodes limit the availability of unaffected

brief background of these two commodities along with a discussion of the events that followed up to the suspension.

2.1.1 *Chana* derivatives suspension

The *chana* derivatives contract was introduced on NCDEX in April 2004, followed by two other pulses, black gram (*urad*) and pigeon pea (*tur*) in July 2004 and April 2005, respectively. However, due to concerns over rising prices, derivatives trading in all three pulses was suspended in 2007 / 2008. Subsequently, while the suspension in *chana* was revoked in December 2008, it continued on black gram and pigeon pea. Among other pulses, a derivatives contract on green gram (*moong*) was introduced in July 2019 but was also suspended in December 2021 amid rising food prices.

India is the world's largest producer and consumer of pulses, with *chana* accounting for over 40 percent of total domestic pulses production and contributing more than 60 percent of global output (Roy and Chandra, 2017). Other major pulses include: pigeon pea (*tur*), green gram (*moong*), black gram (*urad*), red lentil (*masur*), which together constitute 80 percent of India's pulse production. The remaining 20 percent comprises arid legumes such as cowpea (*lobia*), horsegram (*kulthi*) as well as *moath* bean.

In terms of domestic consumption, *chana* is second after *tur*. The bulk of *chana* demand is met through domestic production, with imports contributing only 4-5 percent of net availability. *Chana* is a *rabi* crop which grows during the cool / dry season, as is red lentil (*masur*), while pigeon pea, black gram, green gram and cowpea are grown during the warmer, rainy *kharif* season (Roy and Chandra, 2017).⁸

Between 2003–04 and 2013–14, *chana* production in India rose from 5.72 to 9.53 Million Tonnes (MT).⁹ However, poor monsoon conditions in 2014–15 led to production shortfalls. The resulting demand-supply mismatch caused sharp price increases, with wholesale prices rising from Rs. 3,500 to Rs. 4,400 per quintal between April and June 2015.¹⁰

In response, in September 2015, the government imposed stockholding limits under the Essential Commodities Act (ECA).¹¹ Poor monsoon continued in 2015-16, further lowering expected yields. Thus, the prices continued to increase. By mid-June 2016, wholesale *chana* prices had risen to Rs. 7,000 per quintal. In June 2016, the derivatives market regulator suspended *chana* futures trading as a measure for “abundant” caution.¹² Despite the suspension, prices continued to rise, reaching

commodities that can serve as valid donor units in the synthetic control framework, and hence we do not analyse those episodes in this study.

⁸There are two major agricultural seasons in India: *Kharif* which involves sowing in June while the harvest is ranges from October to January; and *Rabi* for which sowing is typically done from October onwards, while the harvest starts from March onwards.

⁹The agricultural crop year in India is from July to June.

¹⁰1 quintal = 100 kilogram

¹¹The ECA regulates the movement and storage of agricultural produce. The objective is to prevent hoarding and black marketing and ensure access to ‘essential commodities’ by consumers at affordable prices. The central and state governments impose ECA restrictions in the event of price volatility or shortage in food supply.

¹²See: <https://www.sebi.gov.in/media/press-releases/jun-2016/trading-in-chana-futures-on-commodity-derivative-exch-32650.html>

Rs. 8,000 - 8,900 in July and peaking at Rs. 12,000 per quintal by October 2016. Figure B.1 in the appendix shows prices for major pulses prior to the derivatives trading suspension. With bumper production and lower prices, derivatives suspension on *chana* was revoked after about a year, in July 2017.

The second and most recent suspension on *chana* derivatives examined in this paper is the August 2021 suspension. Between 2017-18 and 2020-21, production reached record levels in three of four years, with annual output exceeding 11 million tonnes (MT). However, prices of *chana* and all major pulses saw a sharp rise after March 2020 (Figure B.2). In response, the government relaxed import restrictions on other pulses and released buffer stocks at lower prices in December 2020 and January 2021. While prices stabilized temporarily, they surged again during the April 2021 Delta wave COVID-19 lockdown. It was only from mid-June onwards that prices started showing some decline.

Despite declining prices, on July 2, 2021, the central government imposed stock limits under the ECA on pulses to rein in any potential future price rise owing to the gradual opening up of the country as well as the upcoming festival season. News reports from that time indicate the officials were worried that pulses demand from hotels, restaurants and catering units would increase once the commercial sector opened up.¹³ On August 16, 2021, trading in *chana* derivatives was also suspended by the market regulator. Unlike the 2016 suspension, no official rationale was provided behind the first such commodity derivatives suspension in that year. It is plausible that, similar to the stock limits, the suspension was preemptive, aimed at averting further price increases. Nonetheless, prices continued to rise (Figure B.2). The suspension was subsequently extended by one year in December 2021 and remains in place as of August 2025.

2.1.2 Mustard seed derivatives suspension

Unlike *chana* which saw multiple episodes of suspension since its introduction, mustard seed derivatives contracts, launched in 2003, were suspended for the first time in October 2021. Soybean derivatives, the only other oilseed contract traded in India, were also suspended in December 2021 amid rising prices. Crude palm oil and refined soy oil, both traded on the derivatives platform, were suspended in December 2021.

India ranks fourth globally in oilseeds production. It accounts for about 15-20 percent of the world's oilseeds area, 6-7 percent of vegetable oils production, and 9-10 percent of edible oils consumption. Seven major edible oilseeds are cultivated in India, of which soybean (34 percent), rapeseed-mustard (27 percent) and groundnut (27 percent) accounted for about 88 percent of domestic oilseeds production in 2018-19. Sunflower, sesame, safflower and niger are the remaining four major edible oilseeds. Besides, small quantities of coconut, oil palm, cottonseed and rice bran oil also contribute to domestic production. Amongst the oilseeds, rapeseed-mustard and safflower are *rabi* crops, while soybean, nigerseed and sesame are grown during the *kharif* season. Groundnut

¹³See <https://theprint.in/india/modi-govt-imposes-stock-limit-on-pulses-to-rein-in-surg-ing-prices/689154/>.

and sunflower are grown during both the seasons, though most of the groundnut is grown during the *kharif* season.

While the production of oilseeds in India has grown steadily since 2016-17,¹⁴ it has not kept pace with rising domestic demand for vegetable oils. Hence, over 60 percent of the India's vegetable oil consumption is met through imports (Government of India, 2021-22).¹⁵

The vegetable oil market in India is dominated by palm oil (38 percent), soybean oil (24 percent), mustard oil (16 percent) and sunflower oil (7 percent). The remaining oils that are consumed in the country are cottonseed oil (6 percent) and groundnut oil (5 percent).

Of the four major oils, India meets almost all of its palm oil demand through imports from Indonesia and Malaysia.¹⁶ Over 60 percent of soyoil demand is met through imports from Argentina and Brazil, while more than 90 percent of sunflower oil supply in the country is made available through imports from Ukraine. By contrast, mustard oil demand is met through domestic production. Despite self-sufficiency in mustard, India's heavy reliance on imported vegetable oils means that soaring international prices of any one oil tend to spill over to others.

Figure B.3 plots the domestic and international prices of vegetable oils. With concerns over falling demand due to the COVID-19 pandemic, international prices of all vegetable oils declined beginning March 2020. However in early May 2020, prices began to rise internationally as COVID-19 restrictions began to ease worldwide. Tighter global supplies due to adverse growing conditions in major producing countries had also put pressure on the prices of oilseeds (?).

In August 2021, reports of significantly lower rapeseed production in Canada due to adverse climate conditions impacted its prices. A USDA report on oilseed markets released on August 12, 2021 estimated that with lower production and lower stocks from previous year, rapeseed supplies from Canada were forecasted to decline by more than 20 percent, the lowest level in a decade (USDA, 2021).¹⁷ International prices of rapeseed thus rose sharply during this period.

During periods of high edible oil prices, the Indian government has intervened through trade policy measures and by imposing stock limits under the Essential Commodities Act. In response to rising prices, stock limits were imposed on oilseeds and edible oils in October 2021. Shortly thereafter, on 8 October 2021, the derivatives market regulator suspended trading in mustard seed derivatives.¹⁸ As with *chana*, this suspension was extended by one year in December 2021 and has remained in place since.

¹⁴The oilseed production grew by 43 percent from 2015-16 to 2020-21.

¹⁵India is the world's second largest consumer and largest importer of vegetable oils (15 percent). See: https://www.indiabudget.gov.in/economicsurvey/ebook_es2022/files/basic-html/page265.html

¹⁶Since 2018, in view of the persistently high imports, the government is promoting the production and productivity of oilseeds under the National Food Security Mission: Oilseeds (NFSM-oilseeds) Scheme. For more details, see (Government of India, 2021-22) and <https://www.nfsm.gov.in/StatusPaper/NM00P2018.pdf>

¹⁷Canada represented more than 60 percent of global trade in rapeseed in the previous four years.

¹⁸See: <https://www.sebi.gov.in/media/press-releases/oct-2021/sebi-issues-directions-to-ncdex-regarding-trading-in-53182.html>

3 Data and Empirical Strategy

3.1 Data

Our analysis focuses on oilseeds and pulses. For this purpose, we rely on multiple data sources. The main variable of interest is domestic prices, for which we collect data from the Government of India, Agricultural Market Portal (*Agmarknet*), a centralized database that collects daily information from regulated wholesale markets (*mandis*) across the country.¹⁹ These *mandis* form the primary marketing channel for sale of agricultural produce in India (Aggarwal and Narayanan, 2023).

At the end of each trading day, each *mandi* reports the minimum price, maximum price and modal price for each commodity, along with information on total arrivals to a *mandi* for that day. In total, there are about 3000 such *mandis* in the country that trade over 200 agricultural commodities. For our purposes, we collect daily data on all oilseeds²⁰ as well as pulses²¹ for the analysis from July 2014 to December 2021, which constitutes our main sample period.

To compute a national level wholesale price for each crop, we calculate the arrivals-weighted average of the modal price for major *mandis*. To ensure a representative sample, we include only those *mandis* whose median annual market share is within the top 20 percentile over the sample period for a particular commodity. We further exclude *mandis* that do not report data consistently by considering only those whose median number of reported days for price and arrivals data is also in the top 20th percentile during the sample period.

To determine the accuracy of the national level wholesale price series that we compute,²² we compare it against a national benchmark, the polled price collected by the agri-commodities derivatives exchange, NCDEX. The polled price represents the average of prices quoted by empanelled participants, that consists of value chain participants including traders, processors, and importers / exporters in the major producing centers of a commodity.²³ However, polled prices are not collected for all commodities. The exchange collects this data for only those commodities that are traded on its derivatives platform, and their close substitutes. We compare the polled prices of these ‘select’ commodities with our national level wholesale price series derived from the data based on the *Agmarknet* portal.

While the *Agmarknet* data and NCDEX polled price data provide data for oilseeds and whole pulses, we source wholesale prices for vegetable oils from the Solvent Extractors Association (SEA), the main industry association of vegetable oil processors in India.

We further require data on factors that affect price behaviour. For this purpose, we collect data on annual production, area under production, and yields from 2014-2022 crop years from the website of the Directorate of Economics and Statistics, Ministry of Agriculture, Government of

¹⁹The data from the portal have been used in several papers including ?, Narayanan and Tomar (2023), Aggarwal and Narayanan (2023) and ?.

²⁰These include cottonseed, groundnut seed, soybean, mustard seed, coconut seed, sunflower, niger seed and linseed.

²¹These include pigeonpea (*tur / arhar*), Bengal gram (*chana*), black gram (*urad*), cowpea (*lobia*), green gram (*moong*), chickpea (*kabulichana*), horse gram (*kulthi*) and red lentils (*masur*).

²²In India, no alternative mechanism provides wholesale market spot price information for agricultural commodities.

²³In computing the average, the polling agency trims the data to remove extreme observations.

India. We also obtain annual and monthly exports and imports data from the Ministry of Commerce and Industry web portal. For oilseeds, we obtain global and domestic oil and oilseeds supply and demand data from the monthly oilseeds report published by the United States Department of Agriculture.²⁴ We source monthly international prices for major vegetable oils from the FAO price database, GIEWS.²⁵ Finally, for oilseeds and pulses that trade on the derivatives platform, we obtain open interest, trading volumes and near month futures price data from NCDEX.

3.2 Empirical strategy

Identifying causal impacts when an intervention is implemented at an aggregate level (e.g. national markets) or affects only a limited number of treated units is challenging using the traditional regression based techniques which rely on large samples and multiple instances of the intervention of interest. In such cases, the synthetic control method (SCM), introduced by Abadie and Gardeazabal (2003), and further developed by Abadie *et al.* (2010) and Abadie *et al.* (2015), is particularly useful. The method has gained significant popularity in recent years with applications spanning across disciplines including economics, political sciences, biomedical research. SCM provides a data-driven, systematic approach to detect the causal effects of a policy intervention on a treated unit by constructing a counterfactual based on a weighted average of *untreated* units (referred as the “donor pool”). This counterfactual, known as the synthetic control, represents the path that the treated unit would have followed in the absence of the intervention. By comparing the observed trajectory of the outcome variable for the treated unit post-intervention with that of the synthetic control, one can estimate the causal impact of the intervention, if any.

The methodology is particularly well suited to our study, since the suspension was implemented at the national derivatives market level for each commodity in each episode. Furthermore, the limited number of such suspension episodes constrains the applicability of regression-based techniques. During other suspension episodes (e.g., in 2007–08 and 2021), the suspension was invoked on multiple commodities simultaneously, leaving no unaffected units to serve as valid controls (see Table A.1). Thus, we analyse only three episodes of derivatives suspension as described in Section 2: the 2016 and 2021 *chana* suspensions, and the 2021 mustard seed derivatives suspension. Below, we discuss the implementation of the methodology in detail using *chana* as an example.

3.2.1 Constructing the synthetic control for *chana*

We consider data on all $J + 1$ pulses, including *chana*, indexed by $j = 1, 2, \dots, J + 1$ and denote the treated unit, i.e. the unit affected by derivatives suspension, *chana* as $j = 1$. All other pulses with $j = 2, 3, \dots, J + 1$ constitute the ‘donor pool’, or the set of comparison units whose derivatives market trading status remained unchanged during the episodes we study. The donor

²⁴Data are available at <https://www.fas.usda.gov/data/oilseeds-world-markets-and-trade>.

²⁵In particular, we collect data on Rape oil (Dutch, f.o.b. ex-mill), Groundnut oil (Any origin, c.i.f. Rotterdam), Palm oil (Crude, c.i.f. Rotterdam), Soybean oil (Dutch, f.o.b. ex-mill), Sunflower oil (f.o.b. North West European Ports). The data are available at <https://fpma.apps.fao.org/giews/food-prices/tool/public/#/dataset/international>.

pool consists of eight pulses: cowpea, horsegram, *kabulichana*, *masur*, *moath*, *moong*, *tur*, *urad*. As suggested by Abadie *et al.* (2015), Abadie (2021), we restrict our attention only to pulses and do not include commodities from other commodity groups such as cereals, spices, oilseeds etc in our control basket to ensure that the donor pool units display characteristics similar to the affected unit, *chana*. Further, due to lack of data on production, we exclude two comparison units, cowpea and *kabulichana*, when production is included in the list of predictor variables.

For the June 16, 2016 *chana* derivatives suspension, we analyse the period from July 2014 to June 2017²⁶ which gives us 105 weekly price observations in the pre-intervention period and 54 weekly price observations in the post intervention period. In the case of second intervention announced on August 11, 2021, we analyse the period from July 2019 to December 2021,²⁷ yielding 112 pre-intervention observations, and 19 post-intervention periods.

With the weights on comparison unit restricted to be non-negative, the technique relies on the convex hull condition, as the methodology is predicated on the idea that a combination of unaffected units can be used to approximate the synthetic control for the treated unit. However, if the treated unit is “extreme” in value for a particular variable, especially the outcome variable of interest, the convex hull condition may not be satisfied. A potential way to mitigate this issue, as suggested by Abadie (2021), is to consider the differenced series for the outcome variable, using deviations from the pre-intervention means. In our analysis, the violation of the convex hull condition arises in the 2021 *chana* suspension event, when *chana* prices were lower than most of the remaining pulses (see Appendix Figure B.2). For that event, we use both the methods, i.e. the raw weekly price series and the differenced weekly price series.

We next describe the predictor variables in our analysis. For each unit j , the predictor variables are denoted by $X_{1j}, \dots, X_{kj}$, a vector of dimension $k \times 1$. Each element of this vector represents the values of the pre-intervention characteristics for the j^{th} unit that we aim to match as closely as possible for the treated unit and its synthetic control. The synthetic control is selected based on W^* such that the difference given by $X_1 - X_0 W^*$ is minimized. Here, W^* is the $J \times 1$ vector of optimal weights on each unit in the donor pool. X_1 represents a $k \times 1$ vector containing the pre-intervention characteristics of the treated unit, while X_0 is a $k \times J$ matrix with values of same variables for the donor pool units (Abadie *et al.*, 2015). In essence, for each of the m^{th} variables, $W^* = (w_2^*, \dots, w_{j+1}^*)$ is the set of weights that minimizes:

$$\sum_{m=1}^k \nu_m (X_{1m} - X_{0m} W)^2$$

where ν_m reflects the relative importance of the m^{th} variable in measuring the discrepancy between X_1 and $X_0 W$ (Abadie *et al.*, 2015). This optimization ensures that variables with greater predictive

²⁶The suspension was revoked in July 2017.

²⁷A second suspension was imposed on several other commodities, including a donor pool commodity, *moong* on December 20, 2021. To avoid the potential impacts of the December 2021 suspension, we restrict the analysis to the period ending December 19, 2021.

power for the outcome receive higher weight, so that the synthetic control more closely resembles the key factors driving the outcome of interest. The choice of ν_m thus plays an important role in determining the optimal weights.

We also define a validation period to assess the fit between the synthetic control and the treated unit's outcome variable in the pre-intervention period as in Abadie *et al.* (2015). In that, we divide the pre-intervention period into training and validation periods. We leave three to four months before the intervention as the validation period.²⁸

For our analysis, the vector of pre-intervention characteristics (or predictor variables) consists of variables including yearly production, average net imports, average prices and *mandi* arrivals, each computed for three periods within the pre-intervention period. These variables are critical in determining the demand and supply of the underlying agricultural commodity, and consequently, its prices. For robustness checks, we explore alternative specifications by dropping one predictor variable at a time. Once we obtain the optimal weight vector, W^* , we estimate the counterfactual price series for *chana* as the weighted average of prices of units in the donor pool. To determine the effect of derivatives suspension on prices of *chana*, we analyse the difference between the prices of *chana* (P_{1t}) and its counterfactual in the post intervention period:

$$P_{1t} - \sum_{j=2}^{j+1} w_j^* P_{jt}$$

where P_{jt} represents the weekly prices of the j^{th} unit in the donor pool.

3.2.2 Constructing the synthetic control for mustard

We follow the same strategy for constructing the synthetic control for mustard as we do for *chana*. We use domestic prices of other oils in the donor pool: crude palm oil (CPO), refined soy oil, groundnut oil, sunflower oil, and cottonseed oil. Where data are available, we also use international oil prices as additional donor pool units. In alternative specifications, we include international oil prices as predictor variables; in these cases, we exclude them from the donor pool.

We are able to obtain reliable international price data for mustard oil, groundnut oil, soybean oil, palm oil, and sunflower oil. Given the small size of the oil and oilseeds derivatives market in India, international mustard oil prices are not expected to be affected by the derivatives suspension in India.²⁹ However, given the high correlation of international prices with domestic oil prices, including them as either the predictor variable or as a donor pool unit can improve the fit of the synthetic control in the pre-intervention period.

In contrast to *chana* which saw a derivatives suspension in 2016 that was revoked in 2017 (limiting our pre-intervention period for the 2021 suspension to 2019-2021), for mustard oil we are able to use a longer pre-intervention period for analysis. Our estimation window includes weekly

²⁸This corresponds to 11 and 15 observations for the 2016 and 2021 suspension episodes respectively.

²⁹Being a net importer of most of the vegetable oils, India is a price taker of these commodities. See Section 2.

mustard oil prices from July 2017 to April 2021 (256 observations). We keep May 2021 to October 6, 2021 (22 observations) as the validation period. Finally, the period from October 16, 2021 to December 19, 2021 is the post-intervention window (11 observations).

As predictor variables, we include pre-intervention values of average prices, domestic production (total domestic supply), global production shock measured as year-on-year percentage change in world production and total *mandi* arrivals for units for which data are available.

3.3 Inference

For inference, we rely on the permutation-based methodology proposed by Abadie *et al.* (2010). We conduct “in-space” placebos where we reassign the intervention to each member of the donor pool (potential control unit) and estimate the impacts. A permutation distribution will thus be obtained by iteratively reassigning the treatment to units in the donor pool and estimating “placebo effects” for each iteration against which we evaluate the effect estimated for the true treated unit. The effect of the treatment on the unit affected by the intervention is deemed significant when its magnitude is extreme relative to the placebo effects.

While we use this technique for inference, we recognise the limitation arising out of a small number of units in the donor pool in our sample. We thus also re-run the model using “in-time placebos”, as suggested by Abadie *et al.* (2015), wherein, we apply faux intervention dates during the pre-intervention period and estimate placebo effects to assess the predictive power of the synthetic control. As in “in-space placebos”, a good overlap or at least a similar trend between the synthetic control and observed data would be a validation of the methodology.

3.4 Validity of controls

To ensure the validity of the synthetic control analysis, it is essential that the comparison group (donor pool) satisfies certain conditions. First, the donor pool units must not adopt interventions similar to the one under investigation during the study period. In our case, none of the donor pool units was subjected to the same intervention during the study period: either derivatives trading for these units did not exist, or had been suspended much prior to the study period, or had continued uninterrupted throughout.

Second, none of the donor pool units should have experienced idiosyncratic shocks affecting the outcome of interest during the study period. This condition holds in our context, as evidenced by the stability of price trends for the donor pool commodities relative to the treated unit (see Appendix Figure B.1-B.3). Moreover, other interventions imposed by the government during the study period, such as trade policy restrictions or measures under the Essential Commodities Act, were applied uniformly across all commodities within the respective commodity group (see Section 2).

Third, the donor pool should consist of units with characteristics similar to those of the treated unit to avoid the interpolation risk. We thus restrict the donor pool to commodities within the same group (e.g., pulses for *chana*, vegetable oils for mustard oil). However, this raises concerns regarding

the no-interference assumption, or the Stable Unit Treatment Value Assumption (SUTVA), which requires that the treatment of one unit does not affect the outcomes of other units. To mitigate this issue, we use international prices as control units, as these are less likely to be influenced by derivatives suspensions in India. This is because India’s derivatives market is minuscule relative to global commodity markets. Thus, any policy suspension on Indian derivatives trading is unlikely to influence global prices.³⁰ We use international prices of oils for *mustard* to construct the synthetic control, in addition to the domestic prices of vegetable oils. For *chana*, however, international prices are not available for the pulses produced and consumed in India, and hence we cannot use global prices for constructing its synthetic control.

A related concern involves the use of domestic prices of similar commodities in the donor pool, and the potential for spillover effects from the derivatives suspension. However, given the low price elasticity of *chana*, its substitutability with other pulses is limited (Subramaniam, 2016). Consequently, the likelihood of significant spillover effects is minimal. Besides, both *chana* and mustard are *rabi* crops, whereas most commodities in the donor pool are grown in the *kharif* season. This seasonal distinction reduces the potential for spillovers. The leave-one-out robustness tests strengthen this conclusion by demonstrating that no single donor pool unit drives the results, even if affected by potential spillovers.

4 Results

We begin by presenting our findings for the more recent interventions on mustard and *chana* in October 2021 and August 2021, respectively. We then turn to the results for the earlier *chana* derivatives suspension episode of June 2016.³¹

4.1 Effect of 2021 derivatives suspension on mustard prices

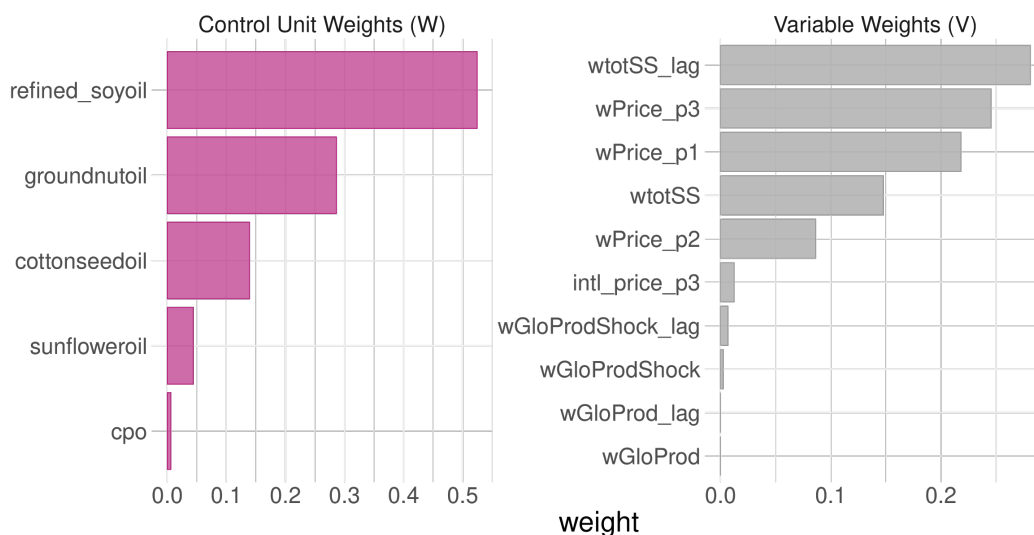
As discussed earlier, to construct the synthetic control of mustard, we use a range of predictor variables. These include pre-intervention domestic prices, domestic and global production, total domestic supply,³² international prices as well as *mandi* arrivals. We also include the year-on-year change in global supply to capture the effect of changes in overall availability on vegetable oil prices. The importance of these predictor variables in tracking the path of the outcome variable (domestic mustard oil prices) is captured in the ν_m vector of weights (Section 3.2.2). We explore different combinations of these predictor variables, and evaluate the ones which give us a good fit, indicated by a close overlap between the pre-intervention synthetic control and the true price series in the test or cross-validation period. The specification that gives us the closest fit is the one with the following predictor variables: pre-intervention prices, total domestic supply, international prices and year on year change in global production.

³⁰Indian commodity exchanges account for less than 3% of global commodity derivatives trading volumes (Aggarwal and Sharma, forthcoming; Government of India, 2018).

³¹For implementation of the methodology, we use the R package *tidysynth* (Dunford, 2021).

³²Total domestic supply includes year-end stocks, net imports along with production

Figure 1 Weights on donor pool and predictor variables for mustard oil



Notes: The graphs show the weights on units in the donor pool and the predictor variables for the first specification. Table A.2 provides the definitions of the predictor variables.

Figure 1 shows the weights obtained for each unit in the donor pool, along with the weights on predictor variables. The most important predictors in this specification are total domestic supply averaged over October 2019 to September 2020, followed by the weekly prices averaged over July 2020 to April 2021 and average weekly prices from July 2017 to June 2019. The two other variables that get a high weight are the total domestic supply averaged over October 2020 to September 2021 and domestic prices between July 2019 and March 2020.

We next discuss the weights obtained for each unit in the donor pool. As shown in Figure 1, the key donor pool units which carry a high weight in the synthetic control for mustard oil under this specification are refined soy oil ($\sim 50\%$) and groundnut oil ($\sim 30\%$). Cottonseed oil gets a weight of less than 15% while sunflower oil gets a weight of less than 5%. We next report the balance table with the pre-intervention characteristics of mustard oil and synthetic mustard oil and the donor pool average in Table 1. Compared to the donor pool average reported in the last column of the table, we observe a lower divergence between the pre-intervention characteristics of mustard oil and synthetic *mustard oil*, particularly for the predictor variables that obtain high weight.

We next examine the effect of the suspension based on the difference between the weekly prices of mustard oil and its synthetic control, as derived from the weights shown in Figure 1. Figure 2 presents the two time series. The grey dotted vertical line marks the end of the estimation period and beginning of the test / validation period. We first assess the fit of the synthetic control with the observed series in the pre-intervention period. While the overlap between actual mustard oil prices and the synthetic control during the estimation period (prior to the grey dotted line) is not perfect, the two series broadly track each other, with the magnitude of their difference remaining small and stable over time. In the validation period, the gap between the two series remains modest

Table 1 Pre-intervention characteristics balance table for *mustard*

This table shows the balance table with pre-intervention characteristics of mustard oil and synthetic mustard oil and the donor pool average. All variables are averaged over the period specified in the parentheses.

Variable	mustard oil	Synthetic mustard oil	Donor sample average
Weekly prices (Jul ‘17 to Jun ‘19)	79.08	77.96	72.44
Weekly prices (Jun ‘19 to Mar ‘20)	85.67	90.17	84.17
Weekly prices (Jun ‘20 to Apr ‘21)	117.24	115.98	113.07
International prices (Jun ‘20 to Apr ‘21)	107.36	129.89	122.00
Total domestic supply (Oct ‘19 to Sep ‘20)	7.99	8.01	8.07
Total domestic supply (Oct ‘20 to Sep ‘21)	8.03	8.00	8.03
Δ YoY global production (Oct ‘19 to Sep ‘20)	1.10	4.94	4.02
Δ YoY global production (Oct ‘20 to Sep ‘21)	3.83	0.27	-2.25
Global production (Oct ‘19 to Sep ‘20)	10.24	9.95	9.88
Global production (Oct ‘20 to Sep ‘19)	10.28	9.95	9.86

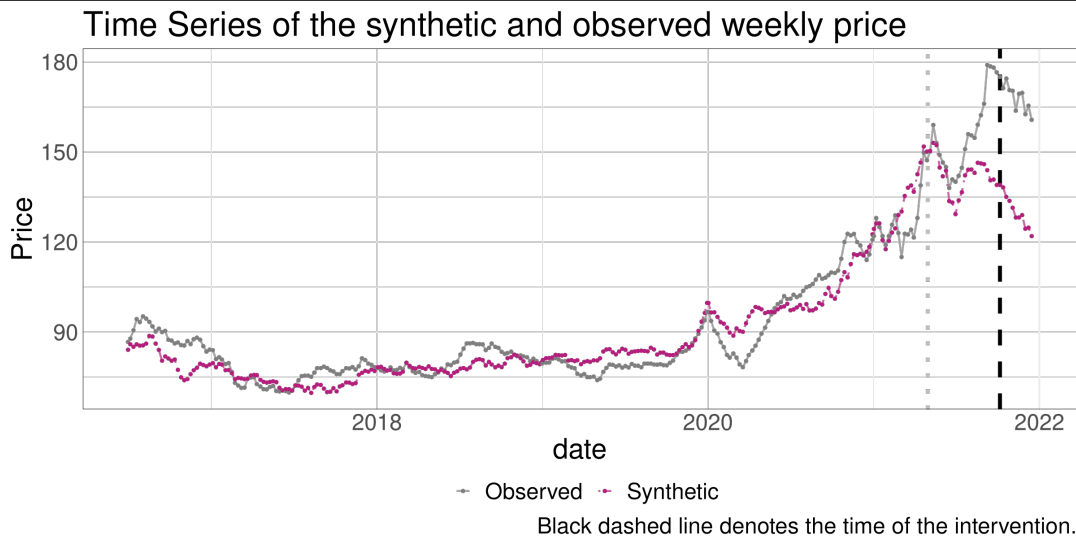
in the first half but widens notably in the second half, around August 2021. This divergence is attributable to the global shock in mustard production following reports of significantly lower than expected production in Canada (Section 2.1.2). This caused a large increase in domestic mustard prices and resulted in the series displaying a behaviour different from the past trend (USDA, 2021). This pattern is also visible in the first graph of Figure B.3, which shows a much steeper rise in mustard oil wholesale prices around August 2021. It is worth noting that although mustard oil prices exhibited a pronounced spike during this period, the synthetic control also partially captured the upward trend, albeit to a lesser extent.

We next examine the trajectory of mustard oil prices and its synthetic control in the post-intervention period. Since mustard oil prices had already risen sharply prior to the intervention due to the global production shock in August 2021, the post-suspension period continues to display a wide divergence between the synthetic control and the observed series. However, we also observe that the prices for both the series show a decline in the post-intervention period. In the absence of a synthetic control analysis, this observed decline would have been attributed to the effect of the derivatives suspension. However, the synthetic control series mirrors the downward trend in observed prices, suggesting that the decline would have occurred even in the absence of the intervention. This evidence goes against the conjecture that the derivatives suspension was responsible for bringing down mustard oil prices.

If derivatives suspension had indeed impacted the prices, we would expect to observe an increase in the prices of the synthetic control relative to the observed series. The absence of an increase in the prices of synthetic control, combined with the same declining trend in both the synthetic control and the observed mustard oil prices, suggests that prices would have followed the same trajectory even in the absence of the suspension. Overall, this provides evidence that the derivatives suspension had no discernible impact on mustard oil prices.

For inference, we conduct the “in-space” and “in-time” placebos as described in Section 3.3. If

Figure 2 Weekly price trends in mustard oil and its synthetic control for the October 2021 suspension



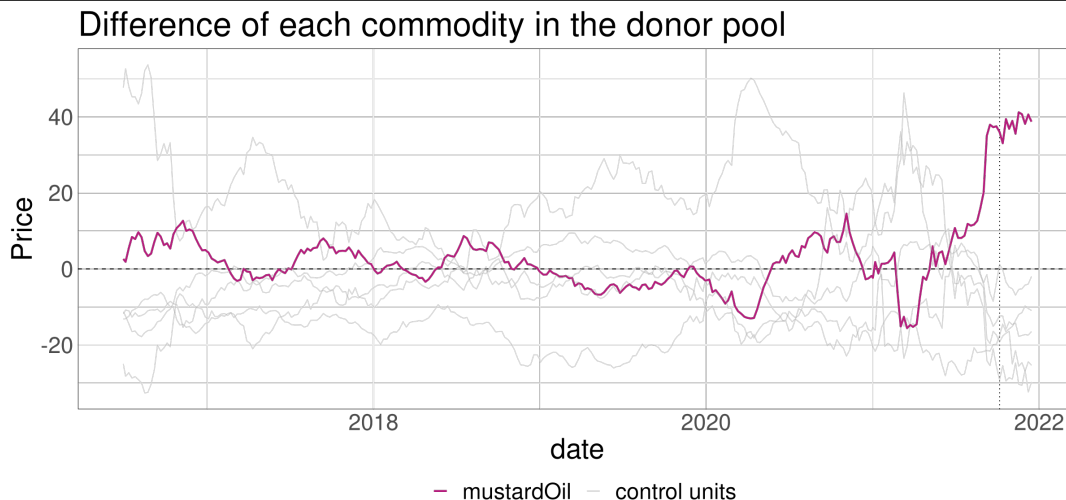
Notes: The graphs show the weekly price trend on mustard oil and its synthetic control for the entire sample period. The grey dotted vertical line in the figure is the end of the estimation period (April 2021) and beginning of the test / validation period. The second vertical line (dashed) is the week in October 2021 when derivatives on mustard oil were suspended.

the magnitude of the difference between the synthetic control and the observed series for the true treated unit (or true intervention date) is unusually large compared to the placebo estimates, we can attribute the observed difference to the intervention. Figure 3 shows the differences in weekly prices between each donor pool unit and its synthetic control, along with the difference between the weekly prices of mustard oil with its synthetic control.

We observe that the price differences of mustard oil and its synthetic control are within the range of price differences generated by the placebo tests, particularly during the pre-intervention period. However, the gap between mustard and its synthetic control widens just prior to the intervention, as noted earlier, due to the global production shock in mustard oil. The effect of this shock persisted in the post-intervention period, as reflected in the sustained size of the price difference even after the suspension. However, if we account for the August 2021 price rise, the post-suspension price difference closely mirrors that of the in-space placebos. Based on this, we infer that the price difference of mustard oil follows the pattern of the donor pool placebos, once again indicating that the lack of any discernible impact of derivatives suspension.

Further, we conduct in-time placebo wherein we reassign the intervention to the middle of the pre-treatment period in October 2019. We use the same set of predictor controls as that of the specification described above (pre-intervention prices, international prices, global production and production shock). The pre-intervention period is set from July 2017 to September 2019 with one month of validation period. Panel A in the Appendix Figure B.5 presents the results. We observe

Figure 3 Weekly price differences based on placebo tests and the treated unit, mustard oil for the October 2021 suspension



Notes: The grey lines in the graph show the differences in weekly prices based on placebo tests by iteratively applying the method on each unit in the donor pool. The superimposed dark pink line shows the difference between the price trend of mustard oil and its synthetic control. The black dashed vertical line (dashed) is the week in October 2021 when derivatives on mustard oil were suspended.

that both the actual and synthetic control time series closely track each other, providing confidence in the predictive power of the synthetic control.

4.2 Effect of 2021 derivatives suspension on *chana* prices

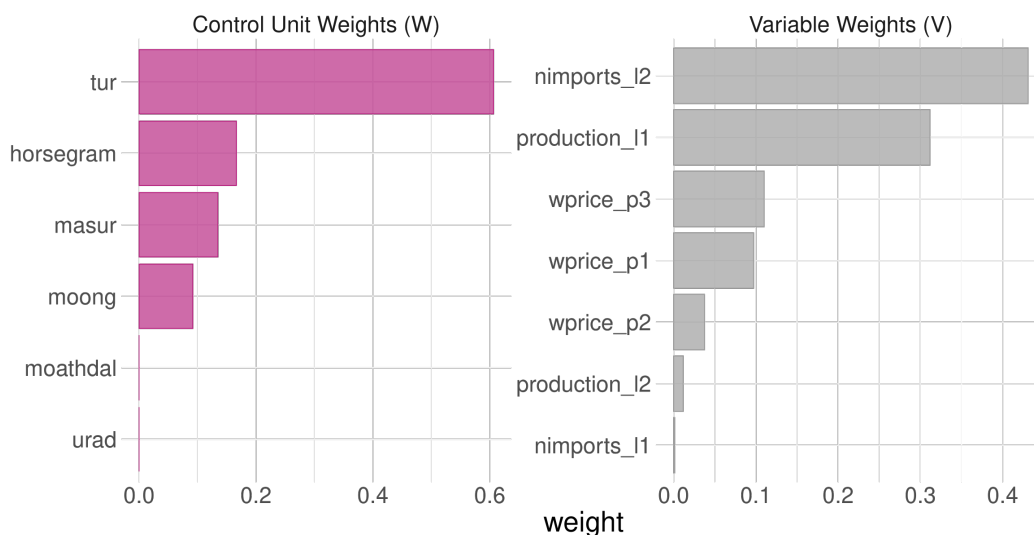
The synthetic control of *chana* is constructed based on the weights (ν_m) selected on the predictor variables, as described in Section 3.2.1. We test different combinations of predictor variables, and identify two specifications that provide a good fit as indicated by a close overlap between the resulting pre-intervention synthetic control and the actual price series in the test / cross-validation period. Both specifications include production, pre-intervention prices, and net imports. The second specification also includes *mandi* arrivals, which improves the overall fit further.

For the 2021 intervention, we report the results based on the differenced series, which provided a better fit compared to the level series.³³ Since both specifications provide similar results, we discuss the results based on the first specification. The results from the second specification, along with other alternatives specifications are discussed in the robustness tests.

For the first specification, the weights assigned to each unit in the donor pool, along with the weights on predictor variables, are shown in Figure 4. The most important predictors include the average monthly net imports data from April to July 2021, production in the crop year 2019-20, demeaned weekly prices from January 2021 to April 2021 and demeaned weekly prices from July

³³The results for the level series are reported in the appendix (Figure B.4). However, the validation period shows a poor overlap between *chana* and its synthetic control. Hence we do not use the level series results for this episode.

Figure 4 Weights on donor pool and predictor variables for *chana*



Notes: The graphs show the weights on units in the donor pool and the predictor variables for the first specification. Table A.3 describes the predictor variables.

2019 to March 2020 (Figure 4). The figure shows that the synthetic price of *chana* is a weighted average of the prices of *tur*, *horsegram*, *masur* and *moong*. Both *tur* and *masur* have higher weights than the other two pulses. The weights on *moath* and *urad* are zero. We also report the balance table with pre-intervention characteristics of *chana* and the synthetic *chana* and the donor pool average in Table 2. In comparison to the donor pool average reported in the last column of the table, we observe a lower divergence between the pre-intervention characteristics of *chana* with that of synthetic *chana*.

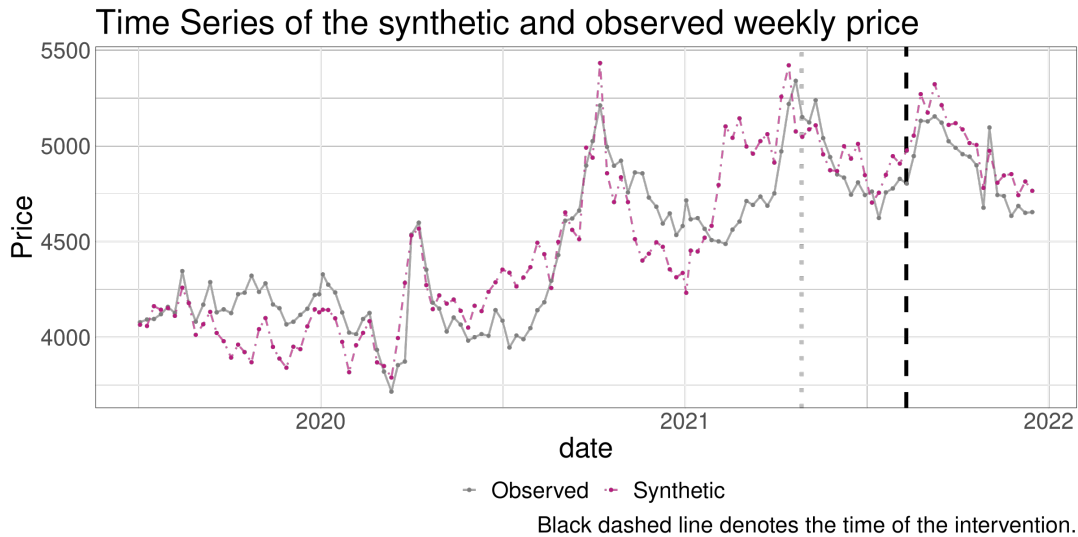
We next examine the effect of the suspension by analysing the difference between the weekly prices of *chana* and its synthetic control, as derived from the weights shown in Figure 4. Figure 5 presents the two time series. The grey dotted vertical line in the figure is the end of the estimation period and the beginning of the test / validation period. During the estimation period (prior the grey dotted line), although the prices of *chana* and its synthetic control do not perfectly overlap, the trend of both the series are similar, and the magnitude of the difference between the two remains small and stable. In the test / validation period, the gap between the two series narrows substantially. This close fit in the validation period provides confidence in the predictive accuracy of the synthetic control based on the selected set of predictor variables.

We next examine the behaviour of the two series in the post-suspension period. We find that both the series closely mimick each other suggesting that the price of *chana* was not impacted by derivatives suspension. The graph indicates that the price behaviour of *chana* would have been similar even in the absence of the suspension. If derivative trading had contributed to price increases, we would expect to see a rise in the synthetic *chana* prices reflecting the counterfactual

Table 2 Pre-intervention characteristics balance table for *chana*

This table shows the balance table with pre-intervention characteristics of *chana* and synthetic *chana* and the donor pool average. Except for annual production variables, the average value of each variable is obtained for the period specified in the parentheses. Demeaned prices imply the differenced price wherein the pre-intervention mean is subtracted from weekly prices.

Variable	<i>Chana</i>	Synthetic <i>chana</i>	Donor sample average
Demeaned Price (Jul '19 to Mar '20)	-250.11	-385.31	-447.25
Demeaned Price (Jun '20 to Dec '20)	346.35	109.21	17.97
Demeaned Price (Jan '21 to Apr '21)	563.22	493.80	440.87
Annual Production (Jul '19 to Jun '20)	9.31	7.63	7.03
Annual Production (Jul '20 to Jun '21)	9.39	7.74	7.15
Monthly Net Imports (Apr '20 to Mar '21)	0.10	0.52	0.32
Monthly Net Imports (Apr '21 to Jul '21)	-0.05	0.08	0.08

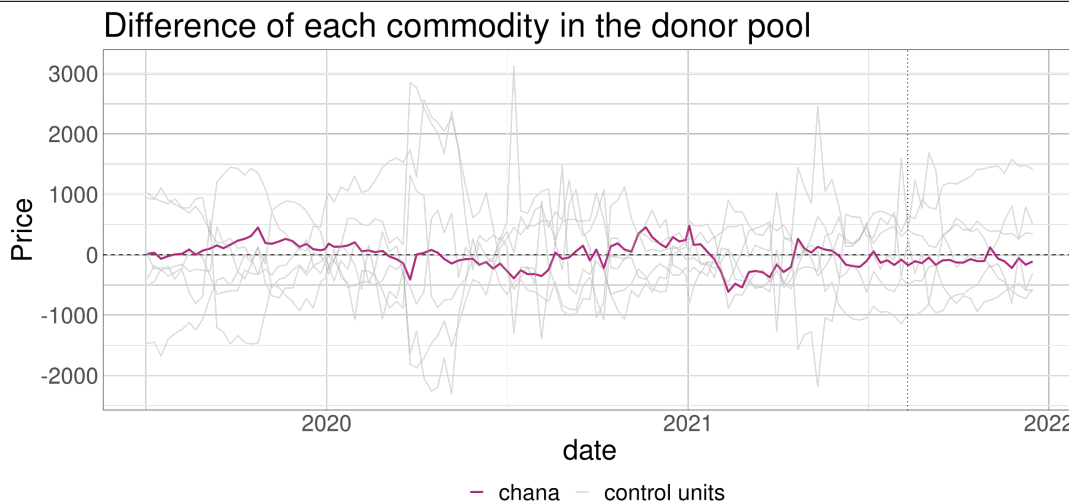
Figure 5 Weekly price trends in *chana* and its synthetic control for the August 2021 suspension

Notes: The graphs show the weekly price trend on *chana* and its synthetic control for the entire sample period. The grey dotted vertical line in the figure is the end of the estimation period and beginning of the test / validation period. The second vertical line (dashed) is the week in August 2021 when derivatives on *chana* were suspended.

path without suspension. However, the findings provide no such evidence. This further supports the conclusion that futures market trading suspension did not impact the spot prices.

To draw inferences, Figure 6 presents the differences in weekly prices between each unit in the donor pool with its synthetic control, alongside the difference between the weekly prices of *chana* with its synthetic control.

Figure 6 Weekly price differences based on placebo tests and the treated unit, *chana* for the August 2021 suspension



Notes: The grey lines in the graph show the differences in weekly prices based on placebo tests by iteratively applying the method on each unit in the donor pool. The superimposed dark pink line shows the difference between price trend in *chana* and its synthetic control. The black dashed vertical line (dashed) is the week in August 2021 when derivatives on *chana* were suspended.

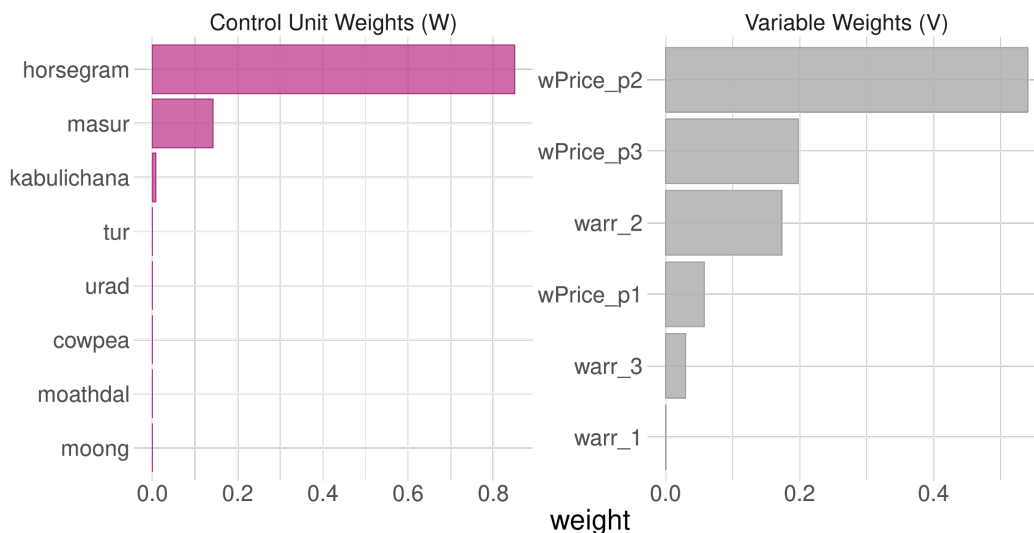
We observe that the price differences between *chana* and its synthetic control are substantially smaller than those observed for any of the donor pool units. Overall, the fit of the synthetic control for *chana* is considerably better than that of the donor pool units. Thus, the “in-space” placebos also indicate that derivatives suspension did not have any impact on weekly prices of *chana*.

We also conduct “in-time” placebo for *chana* by reassigning the intervention date to July 2020, one year prior to the actual intervention date. All other predictor variables are kept unchanged. The second graph in the Appendix Figure B.5 shows a close fit between the observed *chana* weekly price series and its synthetic control, reaffirming the predictive validity of the synthetic control based on the variables used in our main specification.

4.3 Effect of 2016 derivatives suspension on *chana* prices

To construct the synthetic control of *chana* for the June 2016 suspension, we follow the same set of predictor variables used in the August 2021 episode. The specification that provides the closest overall fit includes predictor variables on pre-intervention prices and *mandi* arrivals (Figure 7). The predictor variables assigned the highest weights are weekly prices averaged from April to June

Figure 7 Weights on donor pool and predictor variables for *chana* derivatives suspension in June 2016



Notes: The graphs show the weights on units in the donor pool and the predictor variables for the first specification. Table A.3 describes the predictor variables.

2015, followed by weekly prices averaged across July 2015 and March 2016. Besides, *mandi* arrivals summed over April to June 2015 also emerge as an important predictor variable (Figure 7).

In terms of weights on donor pool units, the figure shows that the synthetic price of *chana* is primarily a weighted average of prices of *horsegram* carrying a weight of over 80%, and the remaining on *masur* and to a small extent *kabulichana*. The weights on the other donor pool units turn out to be zero. We also report the balance table with pre-intervention characteristics of *chana*, its synthetic counterpart and the donor pool average in Table 3. We observe a lower divergence between *chana* and synthetic *chana*, particularly for prices.

The pre-intervention weekly prices of *chana* and its synthetic control show a close overlap (Figure 8). This holds during the test period (from April to June 2016, immediately prior to the suspension) as well. Examining the behaviour of weekly prices in the post intervention period, we find that the synthetic control prices are lower than the observed *chana* prices. The trend in both the series is similar, that of a decline. By March 2017, the two series converge. In terms of the effect of derivatives suspension, the evidence once again points to an absence of impact on prices. The observed price changes appear to have been driven by factors external to those capture in the publicly available data.

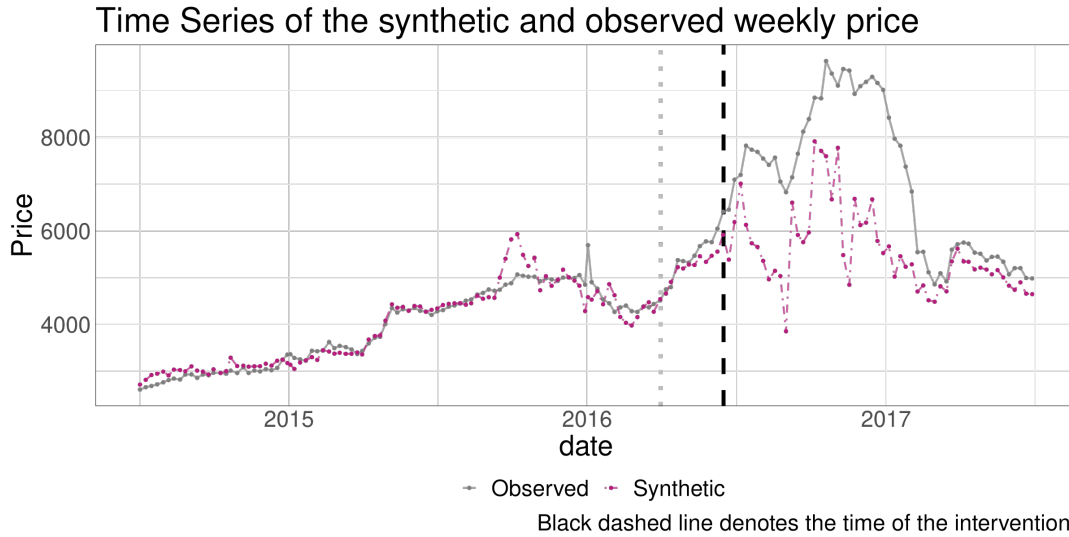
Lastly, we discuss the placebo tests. Figure 9 presents the differences in weekly prices for the “in-space” placebo. The differences for the true treated unit, (*chana*) fall within the range observed for the placebo effects generated by randomly assigning the treatment to donor pool unit. This provides confidence in our inference that *chana* prices were not impacted by the derivatives suspension even in the 2016 episode. The results for the “in-time” placebo for this episode with the intervention date re-assigned to June 2015, are shown in the third panel of Appendix Figure

Table 3 Pre-intervention characteristics balance table for *chana* derivatives suspension in June 2016

This table provides the balance table with pre-intervention characteristics of *chana* and synthetic *chana* and the donor pool average. Mean weekly prices are averaged over the period mentioned in the parentheses. In addition, sum of arrivals during the corresponding period (as indicated in the parentheses) is taken as a proxy for market supply.

Variable	<i>Chana</i>	Synthetic <i>chana</i>	Donor sample average
Weekly prices (Jul '14 to Mar '15)	3157.39	3149.69	4848.15
Weekly prices (Apr '15 to Jun '15)	4026.93	4088.52	5831.77
Weekly prices (Jul '15 to Mar '16)	4832.71	4968.10	6738.87
Log arrivals (Jul '14 to Mar '15)	106.73	88.75	86.58
Log arrivals (Apr '15 to Jun '15)	31.93	27.66	24.37
Log arrivals (Jul '15 to Mar '16)	82.61	68.01	71.45

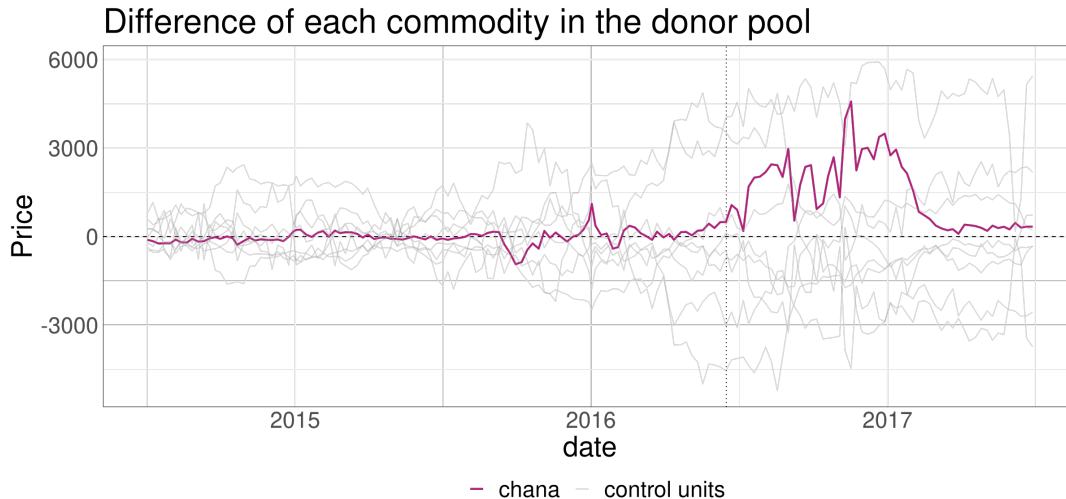
Figure 8 Weekly price trends in *chana* and its synthetic control for the June 2016 suspension



Notes: The graphs show the weekly price trend on *chana* and its synthetic control for the entire sample period. The grey dotted vertical line in the figure is the end of the estimation period and beginning of the test / validation period. The second vertical line (dashed) is the week in June 2016 when derivatives on *chana* were suspended.

B.5. The figure shows a close fit between the observed weekly prices of *chana* and its synthetic control in both the pre- and post-intervention period, validating the predictive performance of the synthetic control in this episode.

Figure 9 Weekly price differences based on placebo tests and the treated unit, *chana* for the June 2016 suspension



Notes: The grey lines in the graph show the differences in weekly prices based on placebo tests by iteratively applying the method on each unit in the donor pool. The superimposed dark pink line shows the difference between price trend in *chana* and its synthetic control. The black dashed vertical line (dashed) is the week in June 2016 when derivatives on *chana* were suspended.

5 Discussion

What explains the findings in the previous section? To understand the results, we examine derivatives traded volumes for each suspended contract. The degree of association between derivatives trading volumes and spot prices offers a few insights.

Figure B.6 plots the daily derivatives trading volumes for each of the three commodities prior the suspension along with their spot prices. A visual inspection of the three graph indicates no direct association between derivatives traded volumes and spot prices. This is particularly evident for mustard derivatives suspension in 2021 and *chana* derivatives suspension in 2016. For these two episodes, we in fact observe that prior to the suspension, derivatives traded volumes for these two episodes do not show any significant increase. In case of *chana* 2021 suspension episode, the derivatives traded volumes display a trend similar to spot prices.

To statistically examine whether derivatives traded volume led spot price changes, we conduct Granger causality test (Appendix Table A.4). In all three episodes, we find no evidence that past changes in trading volumes contained information to forecast future spot price changes.³⁴ The

³⁴We conduct this test only to examine lead-lag relation between the two variables, traded volumes and spot

absence of any statistically significant lead-lag relation between traded volumes and spot prices helps explain why the suspension of trading activity did not impact spot prices.

Further, we note that the size of the domestic derivatives market is smaller than that of the spot market in India. Even for the two commodities analysed in this study, mustard and *chana*, which had relatively high derivatives turnover compared to other agricultural commodities, derivatives market volumes were modest relative to the size of the physical markets. Annual traded volumes represented approximately 78% of total deliverable supply for mustard and 51% for *chana*. The open interest as a proportion of total deliverable supply was even smaller, at around 1% for both commodities (National Commodity & Derivatives Exchange, 2022a,b). The likelihood of disorderly pricing in the spot market due to derivatives trading is typically higher when open interest exceeds deliverable supply (Financial Conduct Authority, 2023). This condition did not apply to either of the commodities analysed in this study.

A natural question that emerges is: what explains the observed rise in spot prices for the three commodities? While a comprehensive analysis of the reasons behind the price rise is beyond the scope of this paper, a brief discussion of the contributing factors is warranted. Supply chain disruptions due to mobility restrictions during the COVID-19 lockdowns, combined with a strong recovery in food demand in the latter half of 2020, likely explain the surge in food prices (Vos *et al.*, 2022). Production shortfalls during this period further exacerbated the increase in food prices, as was evident from the sharp rise in mustard oil prices following Canada’s announcement of a decline in canola production in July-August 2021 (Section 2.1.2). Mustard prices in India were following the trends in world prices, as is also confirmed from the structural break tests. Table A.5 reports the break dates based on the structural break test on domestic and international mustard prices. The break dates on domestic prices closely align with those in international prices. This suggests that domestic prices were moving in line with the international prices of the commodity.

Further, as discussed in Section 2.1.1, the production shortfall in *chana* on account of poor monsoons in 2014-15 and 2015-16 resulted in excess demand and thus high prices. As high prices prevailed for two continuous years, the *kharif* planting season of June-July 2016 saw a sharp increase in pulses acreage. In anticipation of a positive supply shock, the scarcity seen in the last two years thus transformed from a near-famine to a near-feast situation (Subramaniam, 2016). Thus, prices of pulses started declining in early July 2016. The price trends observed during this period thus seem to be primarily responding to the demand-supply mismatches rather than derivatives market operations.

6 Sensitivity tests

In this section, we conduct a range of robustness tests to assess the sensitivity of our main results. We describe these tests below.

prices. Even the rejection of the Granger causality test would not imply that increase in traded volumes was leading to increase in prices, since the two are endogenous.

6.1 Donor pool: Leave one out

In this test, we iteratively re-estimate our baseline model to construct the synthetic control for each of the three episodes by dropping one control unit from the donor pool at a time. This ensures that the results are not driven by any single donor pool unit. Figure B.7 presents the results for all three suspension episodes. The graph shows the trajectory of the observed series (solid black line) alongside its synthetic control from the main specification (blue line). In addition, it shows weekly prices from the synthetic controls in the post-intervention period based on leave-one-out estimates (grey lines). In all the three graphs, we observe a similar behaviour of the leave-one-out synthetic control as that of the baseline model’s synthetic control discussed in the main text. We do not find any series displaying a significantly divergent behaviour in terms of price trends, thus indicating that the results are not sensitive to any particular donor pool unit.

6.2 Alternative specifications

We next analyse the distribution of synthetic controls in the post-intervention period by incorporating different specifications in the model based on variations in predictor variables. The test examines the sensitivity of the results based on changes in predictor variables. Figure B.8 shows the plot for all three suspension episodes. The black line shows the observed price series, while the blue line shows the synthetic control in the post-intervention period for the main specification. The grey lines show the synthetic controls based on alternative specifications. As before, we do not find a substantially different behaviour of the synthetic control series compared to the main specification, providing reassurance that the results are not sensitive to alternative specifications of predictor variables.

7 Conclusion

The recent surge in food prices has renewed interest in the need for, and effectiveness of, policy measures to curb inflation. As in earlier instances of food price rise in India, derivatives suspension has been among the policy tools employed by policymakers in response to food price inflation. This paper examines the impact of such trading suspension in combating food price inflation. Using publicly available data and modern causal inference methods that facilitate better identification, we assess if derivatives trading suspensions contribute to reducing food price inflation.

We study the three most recent episodes of derivative market suspension in India: *chana* in 2016 and 2021 and mustard in 2021. Across these episodes, we find no evidence that derivatives trading suspension affected food prices. The analysis shows that prices of both the commodities, *chana* and mustard, would have followed a similar trend even in the absence of suspensions. Placebo and sensitivity tests re-affirm our findings.

While a comprehensive analysis of the factors driving the observed price trends is beyond the scope of this paper, we find suggestive evidence that mustard oil prices in India were mirroring the international price trends. COVID-19 lockdowns and the gradual opening of the economies

worldwide disrupted the supply chains and impacted food prices globally (Vos *et al.*, 2022). In the case of *chana*, production shortfall and the resulting supply demand mismatch caused the prices to rise in 2015-16 (Subramaniam, 2016). Graphical evidence and Granger causality tests do not show any strong linkages between trading volumes and spot prices for the episodes analysed in this study, further explaining the absence of any impact of trading suspensions on prices.

The advantages of derivatives market in aiding price discovery and risk management have been highlighted in several studies including Cheng and Xiong (2014). Abrupt suspensions adversely affect value chain participants and discourage the development of a liquid derivatives market. Recent news articles and reports suggest that several farmer producer organisations benefit by locking-in prices in derivatives markets (NCDEX, July 2022). Efficient functioning of these markets can bring stability in agricultural markets and also shield farmers from a crash in spot prices in the peak season (?). With increasing climate risk, the need for such a hedging tool for risk management becomes even more important. These considerations, along with the findings of this study emphasise the development of a well-functioning agricultural derivatives market, and avoiding abrupt measures such as trading suspensions during periods of high food inflation that discourage the growth of these markets.

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Appendix A Tables

Table A.1 Agricultural derivatives suspensions in India

Commodity	Suspension year	Commodity	Suspension year
Raw Jute	2005-	Potato	2008, 2014-
Rice	2007-	Sugar	2009-
Wheat	2007, 2021-	<i>Guar</i>	2012
<i>Chana</i>	2008, 2016, 2021-	Mustard seed	2021-
<i>Tur</i>	2007	Moong	2021-
<i>Urad</i>	2007-	Soybean	2021-
Soy oil	2008, 2021-	Paddy	2021-
Rubber	2008-	Crude palm oil	2021-

Table A.2 List of predictor variables used for mustard synthetic control

Mustard 2021: Predictor variables	Name	Description
Weekly prices (Jul ‘17 to Jun ‘19)	wPrice_p1	Average weekly wholesale oil prices from July 2017 to June 2019.
Weekly prices (Jun ‘19 to Mar ‘20)	wPrice_p2	Average weekly wholesale oil prices from June 2019 to March 2020.
Weekly prices (Jun ‘20 to Apr ‘21)	wPrice_p3	Average weekly wholesale oil prices from June 2020 to April 2021.
International prices (Jun ‘20 to Apr ‘21)	intl_price_p3	Average monthly international oil prices from June 2020 to April 2021 based on FAO data.
Total domestic supply (Oct ‘19 to Sep ‘20)	wtotSS_lag	Total domestic supply of all oils from October 2019 to September 2020 based on USDA data
Total domestic supply (Oct ‘20 to Sep ‘21)	wtotSS	Total domestic supply of all oils from October 2019 to September 2020 based on USDA data
Δ YoY global production (Oct ‘19 to Sep ‘20)	wGloProdShock_lag	Year-on-year change in global production for October 2019 to September 2020 based on USDA data
Δ YoY global production (Oct ‘20 to Sep ‘21)	wGloProdShock	Year-on-year change in global production for October 2020 to September 2021 based on USDA data
Global production (Oct ‘19 to Sep ‘20)	wGloProd	Yearly global production for October 2019 to September 2020 based on USDA data
Global production (Oct ‘20 to Sep ‘21)	wGloProd	Yearly global production for October 2020 to September 2021 based on USDA data

Table A.3 List of predictor variables used for *chana* synthetic controls

<i>Chana</i> (2021): Predictor variables	Name	Description
Demeaned prices (Jul '19 to Mar '20)	wPrice_p1	Average weekly differenced price series (with deviations from the pre-intervention mean) July 2019 to March 2020.
Demeaned prices (Jun '20 to Dec '20)	wPrice_p2	Average weekly differenced price series (with deviations from the pre-intervention mean) June 2020 to December 2020.
Demeaned prices (Jun '21 to Apr '21)	wPrice_p3	Average weekly differenced price series (with deviations from the pre-intervention mean) June 2021 to April 2021.
Annual production (Jul '19 to Jun '20)	production_l1	Annual domestic production for the crop year 2019-20
Annual production (Jul '20 to Jun '21)	production_l2	Annual domestic production for the crop year 2020-21
Monthly net imports (Apr '20 to Mar '21)	nimports_l1	Average monthly net imports (difference between exports and imports) from April 2020 to March 2021
Monthly net imports (Apr '21 to Jul '21)	nimports_l2	Average monthly net imports (difference between exports and imports) from April 2021 to July 2021
<i>Chana</i> (2016): Predictor variables	Name	Description
Weekly prices (Jul '14 to Mar '15)	wPrice_p1	Average weekly differenced price series (with deviations from the pre-intervention mean) July 2019 to March 2020.
Weekly prices (Apr '15 to Jun '15)	wPrice_p2	Average weekly differenced price series (with deviations from the pre-intervention mean) June 2020 to December 2020.
Weekly prices (Jul '15 to Mar '16)	wPrice_p3	Average weekly differenced price series (with deviations from the pre-intervention mean) June 2021 to April 2021.
Log arrivals (Jul '14 to Mar '15)	warr_l1	Sum of weekly <i>mandi</i> arrivals from July 2014 to March 2015.
Log arrivals (Apr '15 to Jun '15)	warr_l2	Sum of weekly <i>mandi</i> arrivals from April 2015 to June 2015.
Log arrivals (Jul '15 to Mar '16)	warr_l3	Sum of weekly <i>mandi</i> arrivals from July 2015 to March 2016.

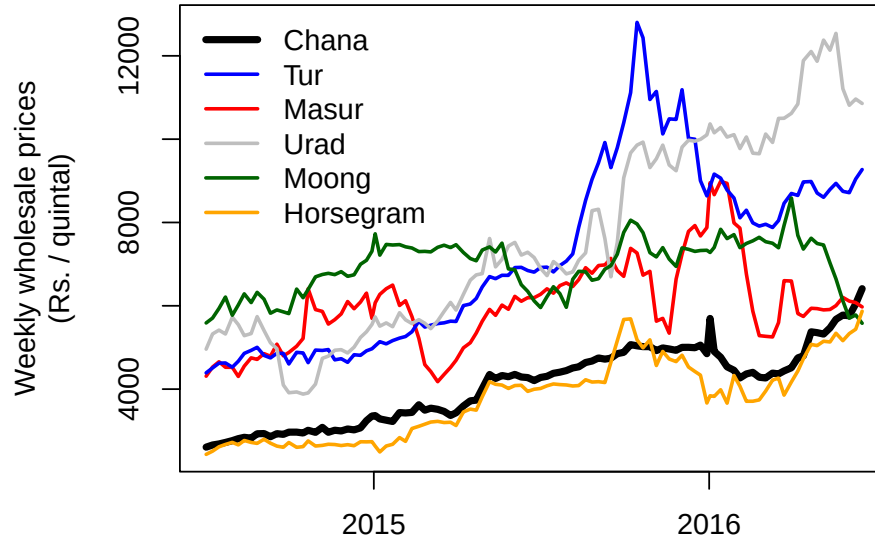
Table A.4 Granger causality test results on *chana* and mustard seed traded volumes and spot price changes prior the suspension

Mustard 2021 suspension		
Granger causality	Order	<i>p</i> -value
Δ Volumes $\rightarrow$ Δ Price	1	0.35
Δ Volumes $\rightarrow$ Δ Price	3	0.42
Δ Volumes $\rightarrow$ Δ Price	5	0.73
<i>Chana</i> 2021 suspension		
Granger causality	Order	<i>p</i> -value
Δ Volumes $\rightarrow$ Δ Price	1	0.47
Δ Volumes $\rightarrow$ Δ Price	3	0.66
Δ Volumes $\rightarrow$ Δ Price	5	0.55
<i>Chana</i> 2016 suspension		
Δ Volumes $\rightarrow$ Δ Price	1	0.96
Δ Volumes $\rightarrow$ Δ Price	3	0.11
Δ Volumes $\rightarrow$ Δ Price	5	0.13

Table A.5 Structural break test on domestic and international mustard prices

Price Series	Break date 1	Break date 2	Break date 3	Break date 4
Mustard oil	2019-11-28	2020-06-11	2020-11-12	2021-04-15
Canola	2019-11-07	2020-06-15	2020-11-29	2021-04-08

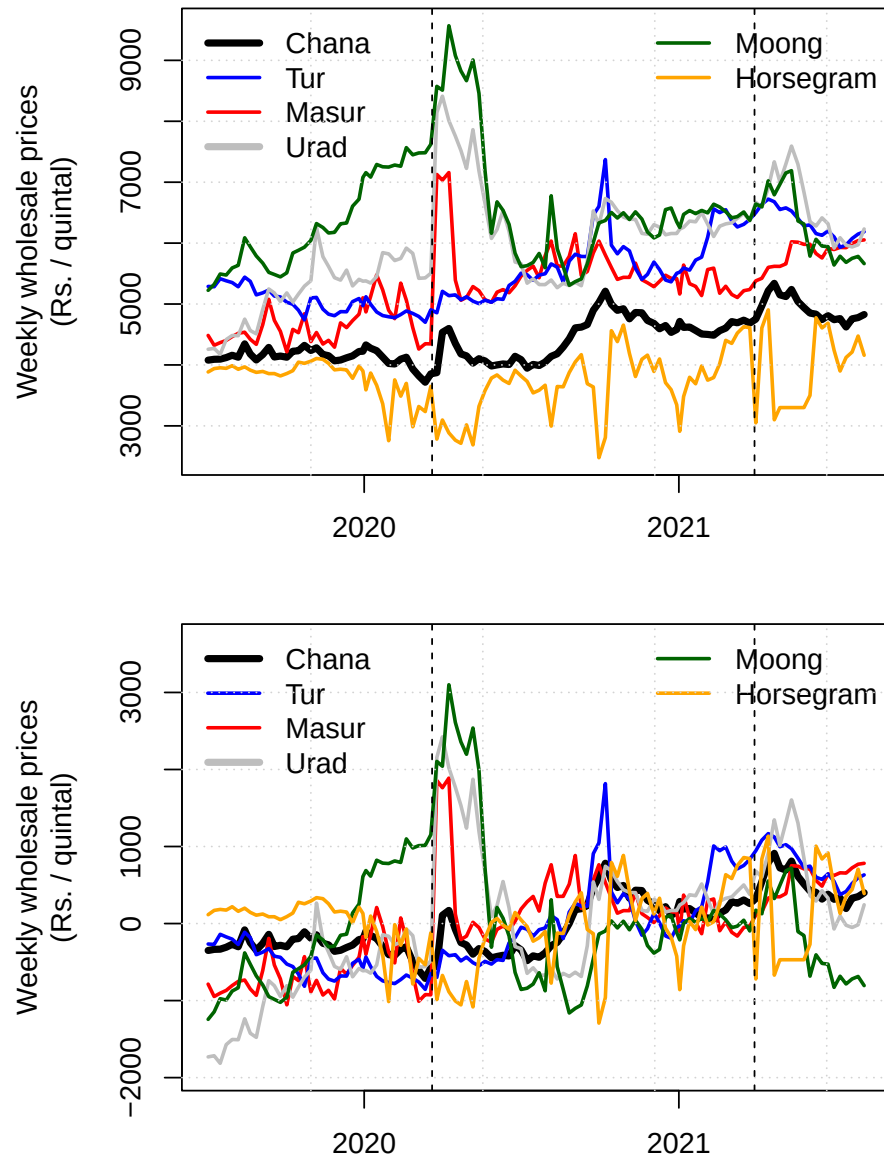
Figure B.1 Prices of major pulses prior the derivatives suspension in June 2016



Notes: The graph shows weekly prices for major pulses from July 2014 to June 2016 based on *mandi* data.

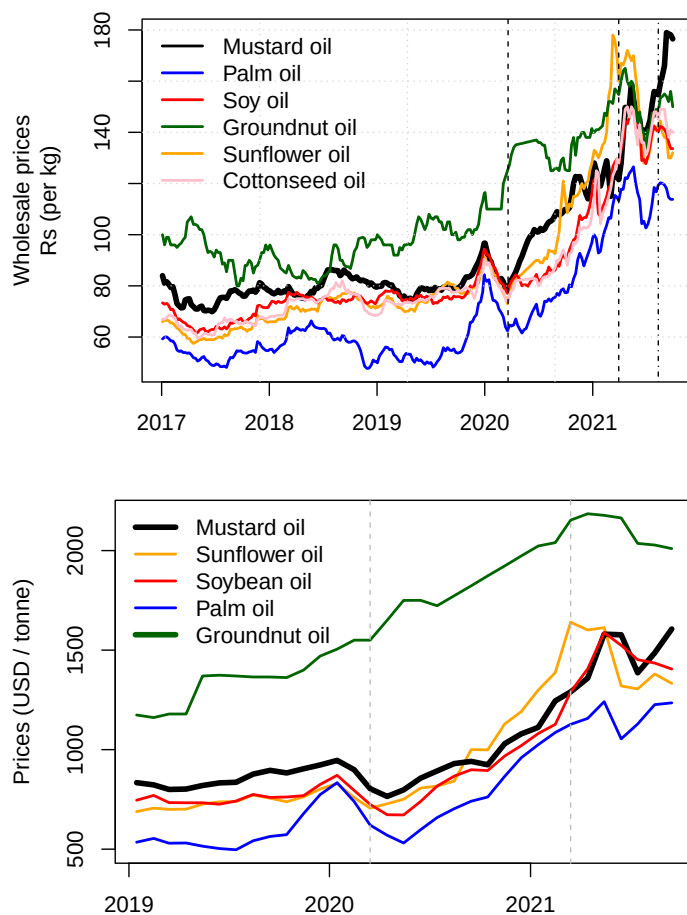
Appendix B Figures

Figure B.2 Prices of major pulses prior the derivatives suspension in August 2021



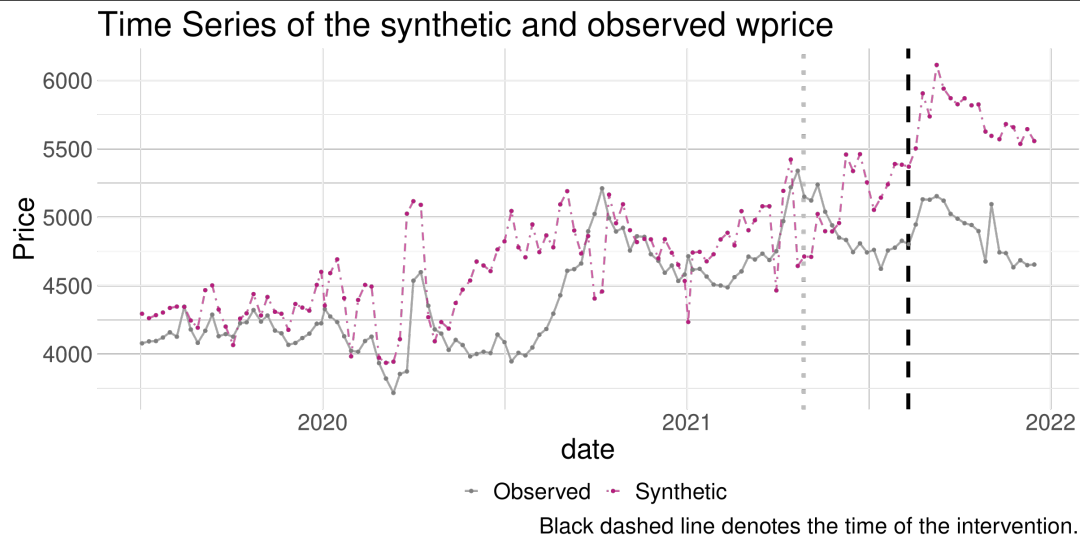
Notes: The first graph shows weekly prices of major pulses from July 2018 to August 2021. Notice that the prices of *chana* are second lowest during this period. The second graph shows the prices in deviations from the pre-intervention mean, as discussed in Section 3.2.1. The first dashed grey line shows the week in March 20, 2020 when a nationwide lockdown was imposed amidst COVID-19, while the second dashed grey line marks the first week in April when the country was hit by the Delta wave of COVID-19.

Figure B.3 Prices of major vegetable oils prior the derivatives suspension in October 2021



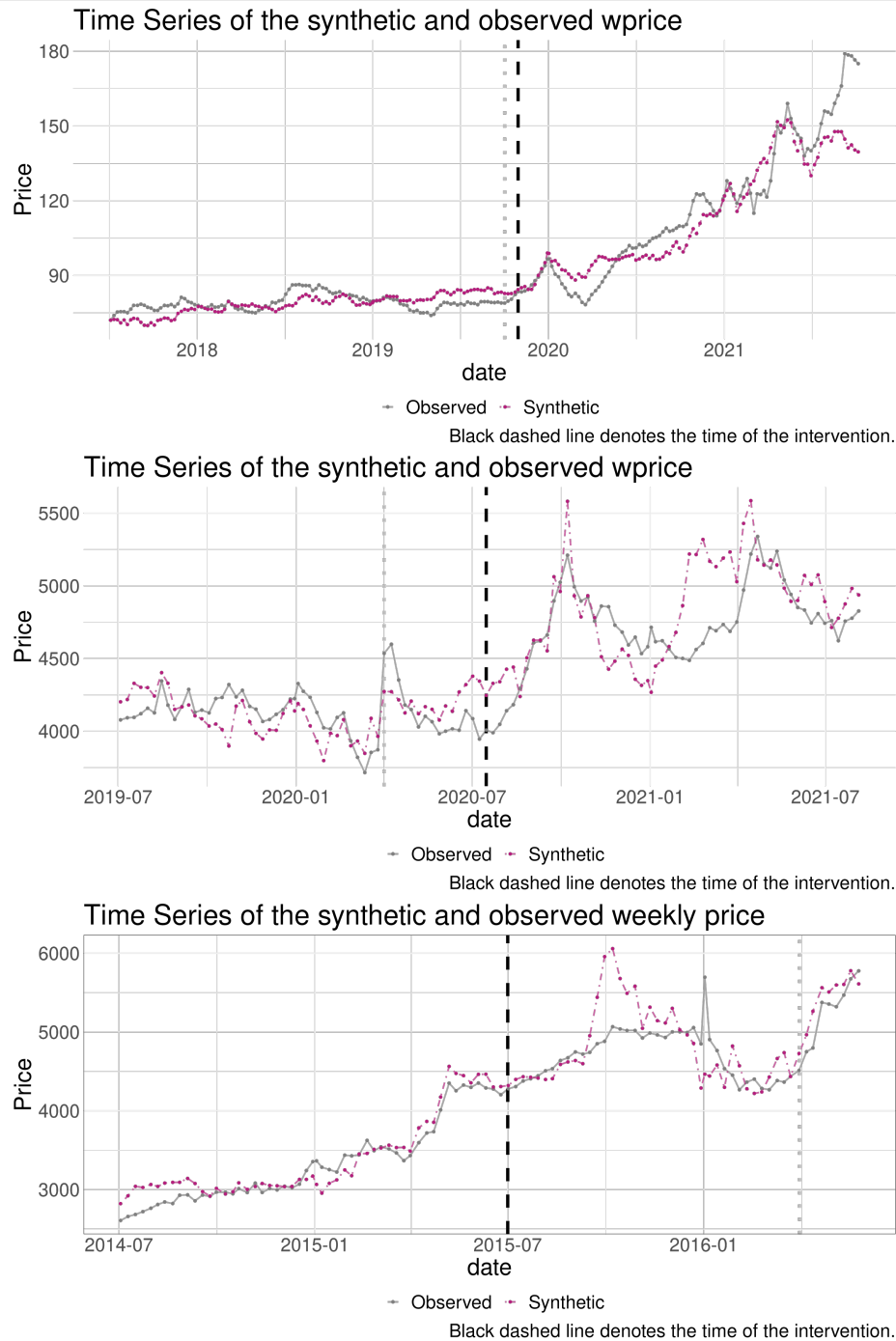
Notes: The first graph shows weekly prices for major oils from January 2019 to September 2021. The second graph shows the monthly international prices of these oils. The first dashed grey line shows the week of March 20, 2020 when a nationwide lockdown was imposed amidst COVID-19, while the second dashed grey line marks the first week of April when the country was hit by the Delta wave of COVID-19.

Figure B.4 Weekly price trends in *chana* and its synthetic control for the August 2021 suspension for the level series



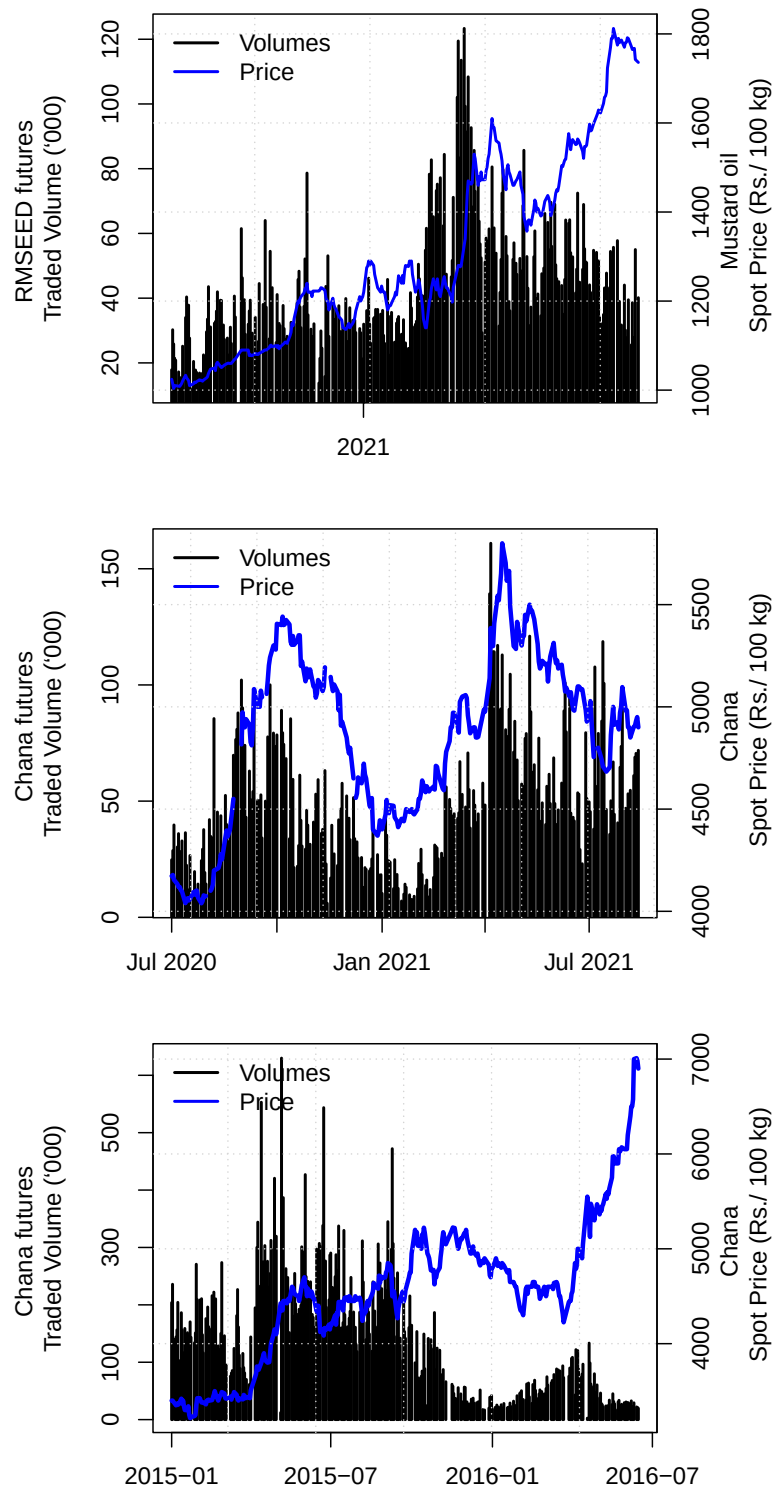
Notes: The graphs show the weekly price trend on *chana* and its synthetic control for the entire sample period based on level series. The grey dotted vertical line in the figure is the end of the estimation period and beginning of the test / validation period. The second vertical line (dashed) is the week in August 2021 when derivatives on *chana* were suspended.

Figure B.5 In-time placebo: Weekly price trends in *chana* and *mustard* with their synthetic controls by reassigning treatment date to a faux date



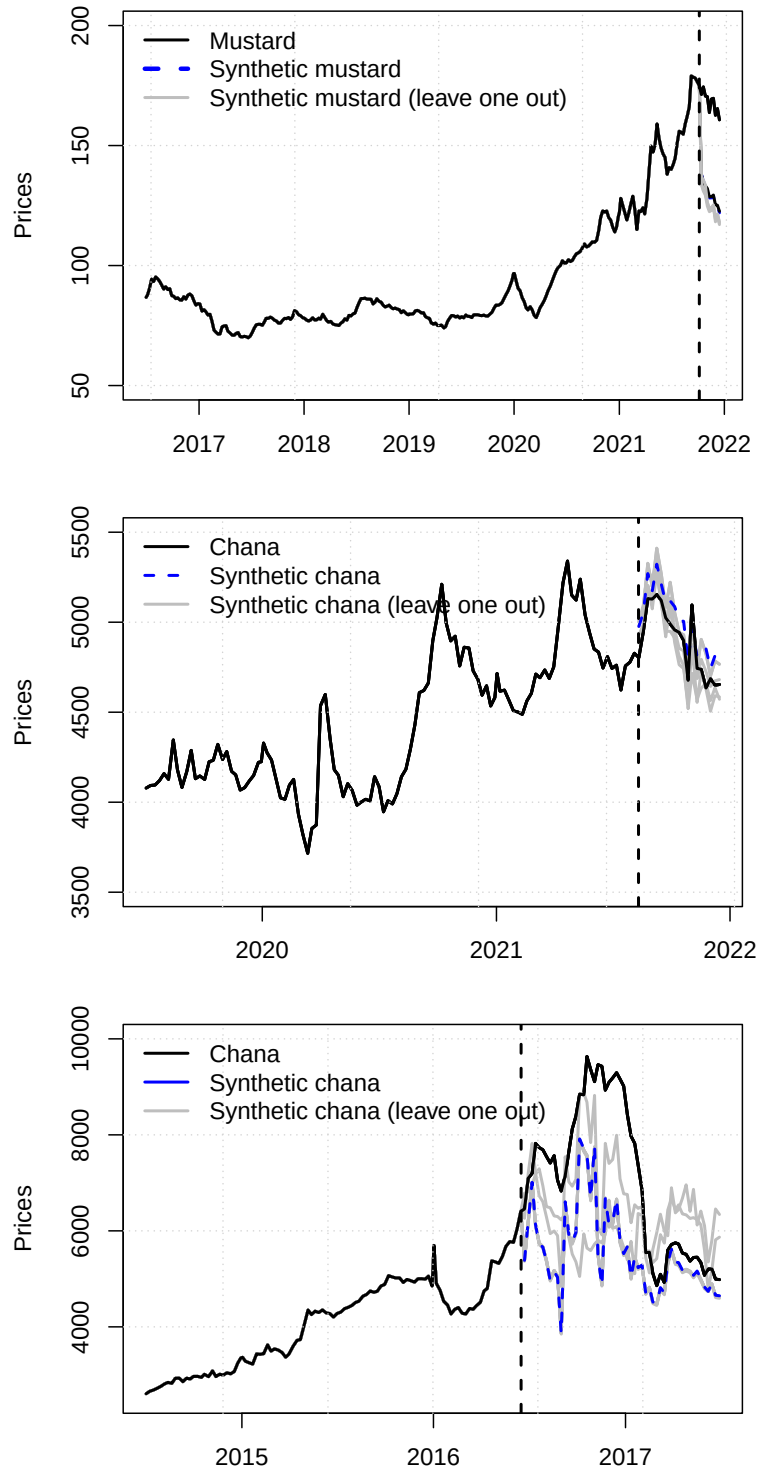
Notes: The first graph shows the in-time placebo for mustard with the re-assigned intervention date as October 2019, while the second and the third graphs show the in-time placebo for *chana* with July 2020 and June 2015 as the re-assigned intervention dates.

Figure B.6 Exchange traded volumes prior to the suspension and spot prices of mustard and *chana*



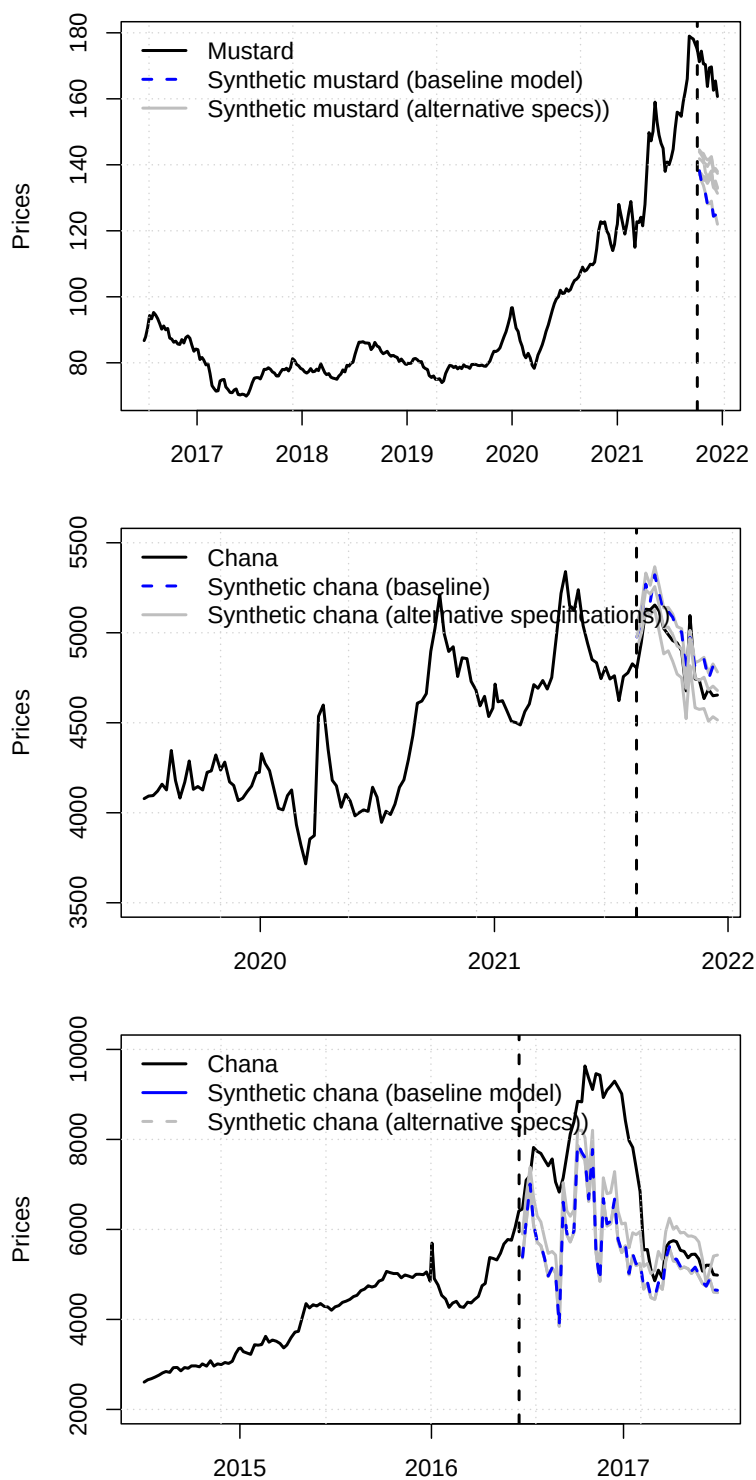
Notes: The first graph shows the exchange traded volumes on futures contract for mustard prior to its suspension in October 2021 along with spot prices. The second and third graphs show the two variables for *chana* prior to the August 2021 suspension and June 2016, respectively.

Figure B.7 Robustness test: Leave-one out distribution of the synthetic control for each episode



Notes: The first graph shows the leave-one-out distribution of the synthetic controls for mustard derivatives suspension of 2021, while the second and the third graphs show the derivatives suspension of *chana* in August 2021 and June 2016, respectively.

Figure B.8 Robustness test: Alternative specifications based distribution of the synthetic control for each episode



Notes: The first graph shows the alternative specifications based distribution of the synthetic controls for mustard derivatives suspension of 2021, while the second and the third graphs show the derivatives suspension of *chana* in August 2021 and June 2016, respectively.